2026-11-08 00:51:42 You

I am a lecturer of “Geology of Ore Deposits” subject.

This subject is called “Ore Deposit Geology”, “Introduction of Mineral Deposits”, “Economic Geology” and etc.

I am going to research the situation of other textbooks and lecture notes of world-famous universities including Cambridge university, and improve the current content of our textbook to make our textbook modern and research-based textbook.

The final goal of improving the textbook is that this textbook must have the modern teaching theory, PBL(Problem-based learning, Project-based learning) teaching.

The current content of our textbook is as following;
1. Basis of Ore Deposits

1.1. Ore

1.1.1 Concept of underground resources

1.1.2 Concept and classification of ores

1.1.3 Mineral composition of ore

1.1.4 Texture and structure of ores

1.1.5 Types of occurrence of valuable components in ores

1.2 Ore body

1.2.1 Concept of ore body and classification of ore bodies

1.2.2 Shapes and attitude of ore bodies

1.3 Ore deposit

1.3.1 Concept of ore deposit

1.3.2 Ore deposit formation system and ore-forming process

1.4 Geotectonic conditions for the formation of ore deposits

2. Magmatic deposits

2.1 Formation and development of magma

2.1.1 Development process of magma

2.1.2 General process of chemical development in magmatic system

2.1.3 Geochemical distribution of elements

2.2 Main types of magmatic deposits

2.2.1 Fractional differentiation deposits

2.2.2 Liquation deposits

2.2.3 Pegmatite deposits

2.2.4 Carbonatite deposits

3. Hydrothermal deposits

3.1 Concept and characteristics of hydrothermal system

3.1.1 Concept of hydrothermal system

3.1.2 Genesis and composition of hydrothermal system

3.1.3 Products of hydrothermal fluid

3.1.4 Classification of hydrothermal deposits corresponding to the types of hydrothermal system

3.2 Magmatic hydrothermal deposits

3.2.1 Porphyritic deposits

3.2.2 Epithermal deposits

3.2.3 Albitite-greisen deposits

3.2.4 Skarn deposits and carbonate-replacement deposits

3.2.5 VHMS deposits

3.2.6 Polymetallic vein deposits

3.3 Orogenic deposits

3.3.1 Orogenic Au deposits

3.3.2 Carlin-type gold deposits

3.3.3 IOCG deposits

3.3.4 Hydrothermal deposits in sedimentary basins

3.3.4.1 Hydrothermal fluids in sedimentary basins

3.3.4.2 MVT deposits

3.3.4.3 SEDEX deposits

3.3.4.4 Red-bed copper deposits

3.3.4.5 Unconformity-related uranium deposits

3.3.4.6 Seafloor hydrothermal deposits

4. Ore deposits formed in sedimentary environments

4.1 Main conditions for the formation of ore deposits formed in sedimentary environments

4.1.1 Concept of ore deposits formed in sedimentary environments

4.1.2 Conditions for the formation of ore deposits formed in sedimentary environments

4.1.3 Oxidation of sulphate deposits (Secondary enrichment)

4.2 Main types of ore deposits formed in sedimentary environments

4.2.1 Residual deposits

4.2.2 Infiltration deposits

5. Sedimentary deposits

5.1 Main conditions for the formation of sedimentary deposits

5.1.1 Concept of sedimentary deposits

5.1.2 Conditions for the formation of sedimentary deposits

5.2 Main types of sedimentary deposits

5.2.1 Placer deposits

5.2.2 Chemical deposits

5.2.3 Biochemical deposits

6. Metamorphic (metamorphosed) deposits

6.1 Main conditions for the formation of metamorphic deposits

6.1.1 Concept of metamorphic deposits

6.1.2 Conditions for the formation of metamorphic deposits

6.2 Main types of metamorphic deposits

6.2.1 Regional metamorphic deposits

6.2.2 Contact metamorphic deposits

6.2.3 Metamorphic hydrothermal deposits

By comparing our content with other textbooks of this subject, please solve the following questions one by one.

- What is the difference and speciality of the content of this subject according to different universities and textbooks related to this subject?

- What is the reason why some of textbooks including Cambridge university do not describe the metamorphic deposits?

- What are textbooks which describe the metamorphic deposits in detail? Please provide the link of such pdf files.

After you answer the above questions, please dive into the writing of the new textbook which can be used for PBL teaching. The total pages of the textbook should be 300+ pages with 150+ images.

- Keeping the original 6 chapters, please provide the improved content of this textbook so that students can get deep understanding of ore deposits. Please note that we should have metamorphic deposits chapter, so I think we should keep the current chapters but should improve sections and subsections. I know that other textbooks deal with mineral exploration and mining, mineral processing and so on. We have other subjects which deal such section, so we only need 6 chapters.

- Please provide the detailed description of each chapter, section and subsection so that I can write wonderful textbook. Here, please suggest the method how to write contents to implement the theory of PBL teaching.

- Please suggest the images to be added to textbook to show students the principle and ore-forming process of each deposit, distribution of ore deposits clearly.

- We have to insert the thinking problems in proper places of each subsection to make students deepen the theory. So, please suggest the thinking problems and answers. Thinking problems will be one of the means of PBL teaching.

- We have to provide 15 review tasks at the end of each chapter. Review tasks must be made to give students the deep knowledge of each chapter, and to make students research the real and useful problems to find ore deposits.

Please list the title, description of each image to be added corresponding to each subsection.

Please note that the original textbook was written by referencing the following textbooks.

Ore Deposit Geology, Cambridge University Press, 2013

The World of Mineral Deposits, 2020

Introduction to Mineral Deposits Geology, 2022

Introduction to Ore-forming Process, 2005

Geology and Mineral Resources, 2025

矿床学，2011

Please search other references and list them so that I can download and research them.

ChatGPT gpt-6-thinking

Modernizing the Textbook: Ore Deposit Geology

A research-based, 300+ page textbook designed for PBL teaching

Your proposed modernization is an excellent opportunity to develop more than an updated textbook. I recommend developing a **genetic, process-oriented, evidence-based textbook** in which students learn to explain why, where, when, and how mineral deposits form, rather than simply memorize the characteristics and classifications of different deposit types.

The most important change should be the teaching philosophy:

- **Traditional approach:** Introduce a deposit type → describe its characteristics → list its minerals → explain its geological setting.

- **Modern approach:** Present a geological problem → introduce geological evidence → ask students to develop competing genetic hypotheses → investigate geological processes → evaluate evidence → construct and defend a deposit model.

For example, instead of beginning a section with the definition of porphyry copper deposits, students could be presented with a geological cross-section containing an intrusive complex, alteration zones, vein networks, and copper mineralization. They would investigate the evidence and determine how the deposit formed.

This approach can be applied to all six existing chapters without changing the overall six-chapter structure.

I would establish five principles for the new textbook:

1. **Genetic understanding:** Explain the physical, chemical, geological, and tectonic processes responsible for ore formation.
2. **Evidence-based learning:** Use geological maps, cross-sections, mineral assemblages, geochemical data, isotope data, fluid inclusions, and published case studies.
3. **Problem-based learning (PBL):** Introduce geological questions and problems throughout every subsection.
4. **Research-based learning:** Connect textbook explanations with recent scientific discoveries and debates.
5. **Integrated geological thinking:** Help students understand relationships between deposit types, geological environments, and the evolution of Earth's crust.

One important recommendation: retain the six chapters, but do not necessarily retain the existing classification system unchanged. Some deposit types currently appear under more than one genetic category, while some important modern deposit groups are missing. The revised textbook should distinguish the *process responsible for mineralization* from the *geological environment in which mineralization occurs*.

Part I. International comparison of textbooks and university courses

1. What are the differences between international textbooks?

There is no universally accepted structure for teaching Ore Deposit Geology. Different authors organize the subject according to their scientific philosophy, educational objectives, and preferred classification of mineral deposits.

My comparison identifies five major approaches.

1.1 Comparison of important textbooks

Textbook	Main approach	Special characteristics
Ridley (2013), *Ore Deposit Geology*	Geological processes and environments	Strong connection between deposit types, petrology, geochemistry, hydrology and tectonics
Robb (2020), *Introduction to Ore-Forming Processes*, 2nd ed.	Ore-forming processes	Excellent explanation of physical and chemical mechanisms, fluid evolution and global metallogeny
Neukirchen & Ries (2020), *The World of Mineral Deposits*	Accessible global overview	Many colour illustrations, examples of important deposits, commodities and their distribution
Pohl (2020), *Economic Geology: Principles and Practice*, 2nd ed.	Integrated metallogenesis	Explicit treatment of metamorphic, metamorphosed and metamorphogenic deposits
Edwards & Atkinson (1986), *Ore Deposit Geology and its Influence on Mineral Exploration*	Deposit characteristics and practical applications	Includes a dedicated chapter on ores formed by metamorphism
Olade (2022/2023), *Introduction to Mineral Deposits Geology*	Broad undergraduate introduction	Includes a separate chapter on metamorphism-related mineral deposits, as well as exploration, mining and economics
Zhai, Yao & Cai (2011), *矿床学*, 3rd ed.	Traditional genetic classification	Includes a dedicated metamorphic deposits chapter and regional metallogeny
Upadhyay (2025), *Geology and Mineral Resources*	Broad geology and mineral resources	Connects mineral deposits with general geology, mineral resources, exploration and resource management

The following are the most important differences.

A. Cambridge University Press – John Ridley (2013)

Ore Deposit Geology

John Ridley, Cambridge University Press, 2013, 409 pages.

Official Cambridge book page

Detailed table of contents

This book is particularly valuable as a structural reference for your textbook.

Its six main chapters are:

1. What is an ore deposit?
2. Magmatic ore deposits.
3. Hydrothermal ore deposits I: magmatic and orogenic environments.
4. Hydrothermal ore deposits II: sedimentary environments.
5. Ore deposits formed in sedimentary environments.
6. Supergene ores and supergene overprinting of ores.

Its strongest characteristics are:

- Deposit descriptions are connected to geological environments.
- Petrological, geochemical, hydrological and tectonic principles are integrated.
- It uses geological examples and explanatory illustrations.
- It includes problems, exercises, review questions and further reading.
- It emphasizes the interpretation and application of geological theory.

The publisher specifically describes the book as a connection between deposit descriptions and the geological processes responsible for their formation. Cambridge University Press Ore Deposit Geology

My recommendation: Use Ridley as the principal reference for the overall organization and academic level of your textbook, but improve it by adding a more comprehensive metamorphic chapter, more explicit PBL activities, and more opportunities for students to interpret actual geological datasets.

B. Laurence Robb (2020)

Introduction to Ore-Forming Processes

Laurence Robb, 2nd edition, Wiley-Blackwell, 2020, 496 pages.

Official publisher page

This is one of the most important scientific references for your proposed textbook.

Its distinctive feature is that it explains *how mineral deposits form*, rather than simply classifying deposits.

Important subjects include:

- Partial melting and fractional crystallization.
- Magma contamination and sulfide saturation.
- Liquid immiscibility.
- Metal partitioning between minerals, melts and fluids.
- Magmatic-hydrothermal processes.
- Fluid sources, including metamorphic water.
- Metal transport through hydrothermal fluids.
- Hydrothermal alteration.
- Porphyry, epithermal, skarn, VMS and sediment-hosted deposits.
- Supergene and sedimentary processes.
- Global tectonics and metallogeny.

Particularly important for your textbook is Chapter 3, which discusses metamorphic fluids and includes orogenic gold deposits associated with aqueous-carbonic metamorphic fluids.

My recommendation: Use Robb as the principal scientific reference for explaining the mechanisms of ore formation in Chapters 2 and 3. It is especially useful for adding quantitative explanations, geochemical reasoning and research-based problems.

C. The World of Mineral Deposits (2020)

The World of Mineral Deposits

Florian Neukirchen and Gunnar Ries, Springer, 2020, 371 pages.

Official Springer book page

This book is particularly valuable for visual communication.

Its major sections cover:

- Introduction to mineral resources.
- The world of metals.

- Magmatic deposits.
- Hydrothermal deposits.
- Deposits formed by sedimentation and weathering.
- Fossil fuels.
- Industrial minerals and rocks.

The book contains 313 colour illustrations, according to its publisher. Springer Nature Link Florian Neukirchen

Its major strengths are accessible explanations, global examples and attractive geological illustrations.

My recommendation: Use this book as an important reference for the design of your figures, global distribution maps, geological cross-sections and introductory explanations. However, your textbook should go deeper into genetic mechanisms and the interpretation of research evidence.

D. Walter L. Pohl (2020)

Economic Geology: Principles and Practice

Walter L. Pohl, 2nd edition, 2020.

Author's official table of contents

Wiley book page

This is particularly important for your decision to retain metamorphic deposits.

Its first chapter explicitly includes:

- 1.1 Magmatic Ore Formation Systems
- 1.2 Supergene Ore Formation Systems
- 1.3 Sedimentary Ore Formation Systems
- 1.4 Diagenetic Ore Formation Systems
- 1.5 Metamorphic and Metamorphosed Ore Deposits
- 1.6 Metamorphogenic Ore Formation Systems
- 1.7 Metallogeny: Ore Deposit Formation in Space and Time
- 1.8 Genetic Classification of Ore and Mineral Deposits.

Contents Wiley Online Library

My recommendation: Use Pohl as the principal reference for the classification and conceptual framework of your new Chapter 6.

1.2 Differences between international university courses

Textbooks are not the only useful references. Actual university course structures provide important information about what students are expected to understand.

Here are some examples.

University/course	Important educational characteristics
University of Geneva – Ore Deposits Introductory Course	Organizes deposits according to geological context and processes; includes separate material on metamorphic rocks and orogenic gold
IISER Mohali – Economic Geology	Emphasizes magmatic processes, hydrothermal fluids, sedimentary basin fluids and global tectonics
IISER Bhopal – Economic Geology	Emphasizes genetic processes, mineral resources, mineral exploration and metamorphic enrichment
University of Oslo – Ore-Forming Processes	Connects geological theory with mineralogical, geochemical, microscopic and field observations
University of Georgia – Economic Geology	Emphasizes interpretation at multiple geological scales, from mineral grains to outcrops and regional settings

Sources:

- University of Geneva – Ore Deposits Introductory Course
- IISER Mohali – Economic Geology
- IISER Bhopal – Economic Geology
- University of Oslo – Ore-Forming Processes
- University of Georgia – Economic Geology

An especially useful example is the University of Geneva course. It has a specific section entitled *Ore deposits in metamorphic rocks and orogenic gold deposits*, while other subjects are organized around magmatic intrusions, volcanic environments, sedimentary basins and surface processes. Ore Deposits Introductory Course iisermohali.ac.in bulletin.uga.edu Universitetet i Oslo

1.3 What should we learn from these differences?

I identify four major approaches.

Approach	Main question	Educational advantage
Genetic/process-based	How did the deposit form?	Deep understanding of ore-forming mechanisms
Geological environment-based	Where did it form?	Understanding of tectonic and geological controls
Deposit-type-based	What are the characteristics of this deposit?	Ability to identify and classify deposits
Commodity-based	Which valuable elements occur and where?	Understanding of mineral resources and their distribution

I recommend combining the first three approaches.

The fourth approach should be used selectively in case studies, rather than becoming the main organizational principle, because your students have other subjects dealing with mineral resources, exploration and mining.

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2. Why do some textbooks, including Cambridge University Press (2013), not have a separate chapter on metamorphic deposits?

This is an important scientific and educational question.

First, there is a clarification: *Ore Deposit Geology* (2013) is published by Cambridge University Press, but its author, John Ridley, was affiliated with Colorado State University. It should not be interpreted as an official textbook written by the University of Cambridge itself. Cambridge University Press Cambridge University Press

Second, the absence of a separate metamorphic chapter does not mean that metamorphic processes are unimportant.

There are several scientific reasons.

2.1 Metamorphism is frequently considered a modifying process rather than an independent ore-forming process

Consider the following geological situation.

An original sulfide deposit may have formed through submarine hydrothermal activity.

Later, the deposit may have experienced:

1. Burial.
2. Increasing temperature and pressure.
3. Regional metamorphism.
4. Deformation and folding.
5. Recrystallization of sulfide minerals.
6. Remobilization and redistribution of metals.

The original deposit was not necessarily formed by metamorphism. Rather, metamorphism modified an already existing deposit.

Therefore, some authors classify the deposit according to its original genetic process and discuss metamorphism as a subsequent geological event.

2.2 There are three scientifically different concepts

This distinction should become one of the most important theoretical components of your new Chapter 6.

A. Metamorphic deposits

Deposits or useful mineral concentrations formed through metamorphic reactions, mineral growth, recrystallization or metamorphic differentiation.

Examples include certain graphite, kyanite, sillimanite and other metamorphic industrial-mineral deposits.

B. Metamorphosed deposits

Deposits formed by an earlier geological process and subsequently modified by metamorphism.

Examples include metamorphosed VMS deposits, metamorphosed stratiform sulfide deposits and metamorphosed nickel sulfide deposits.

C. Metamorphogenic deposits

Deposits formed or substantially upgraded through the mobilization, transport and precipitation of metals during metamorphic processes.

Examples include synmetamorphic gold mineralization and certain metamorphic-fluid-related tungsten deposits.

These terms are not used identically by every author, so your textbook should explicitly state the terminology and classification convention it adopts.

The scientific basis for this distinction is discussed in the Society of Economic Geologists' *Metamorphosed and Metamorphogenic Ore Deposits*. Google Books researchonline.jcu.edu.au segweb.org

2.3 Why does Ridley not have a separate metamorphic chapter?

Ridley's book organizes deposits into six chapters according to major ore-forming environments and processes.

In particular, it places magmatic and orogenic environments together in its hydrothermal chapter.

This is a reasonable classification because some orogenic gold deposits are associated with metamorphic fluids, while hydrothermal processes are central to their formation.

Consequently, metamorphic-fluid-related mineralization can be discussed within a hydrothermal classification without creating a separate chapter.

There are also practical textbook-design reasons:

- The author must balance the breadth and depth of the subject.
- Some deposit types involve several overlapping geological processes.
- A separate metamorphic chapter could repeat material already covered under hydrothermal deposits.
- A textbook designed for a particular undergraduate course may prioritize the most commonly taught metallic deposit types.

These are reasonable interpretations of the book's structure, not an explicit statement by Ridley that metamorphic deposits are unimportant.

2.4 My recommendation for your textbook

(Keep Chapter 6)

I strongly recommend retaining a dedicated chapter on metamorphic and metamorphosed deposits.

There are three reasons:

1. Metamorphism can significantly modify the mineralogy, texture, structure, grade and geometry of existing deposits.
2. Metamorphic fluids can transport and concentrate economically important metals.
3. Students need to understand the distinction between primary mineralization, metamorphic modification and new mineralization associated with metamorphic processes.

The chapter should not simply describe metamorphic deposits as another classification category. It should explain the relationship between metamorphism and the entire evolution of mineral deposits.

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3. Textbooks and scientific references that describe metamorphic deposits in detail

I would prioritize the following references.

3.1 The most important references

1. Metamorphosed and Metamorphogenic Ore Deposits

F. M. Vokes, B. Marshall and P. G. Spry, editors. Society of Economic Geologists, Reviews in Economic Geology, Volume 11, 1998/2000, 310 pages.

(Highest priority)

This is the most directly relevant specialized reference. It covers metamorphic remobilization, metamorphic fluids, metamorphosed massive sulfide deposits, nickel sulfides, tungsten mineralization and synmetamorphic gold.

Official SEG PDF edition (paid PDF)

Free PDF containing contents and publication information

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2. Ore Deposit Geology and its Influence on Mineral Exploration

Richard Edwards and Keith Atkinson, 1986.

Chapter 11: *Ores Formed by Metamorphism*, pages 337–369.

This is an especially useful reference for the organization of a metamorphic deposits chapter.

Springer book and chapter access

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3. Economic Geology: Principles and Practice

Walter L. Pohl, 2nd edition, 2020.

Section 1.5: Metamorphic and Metamorphosed Ore Deposits.

Section 1.6: Metamorphogenic Ore Formation Systems.

Detailed contents

Publisher access

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4. Introduction to Mineral Deposits Geology

Moses Olade, 2022 (some catalogues record 2023).

Chapter 6: Metamorphism-Related Mineral Deposits.

This book is useful for an undergraduate-level explanation, although it also covers exploration, mining and mineral economics.

Google Books bibliographic information

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5. 矿床学 (Mineral Deposits)

翟裕生、姚书振、蔡克勤主编，地质出版社，2011，第三版。

Chapter 12: 变质矿床 (Metamorphic Deposits).

Particularly useful for understanding the traditional genetic classification and Chinese geological case studies.

Book details and table of contents

The SEG volume deserves particular attention: its contents include chapters on metamorphic fluids, regional metamorphic remobilization, metamorphosed massive sulfides, komatiite-hosted nickel sulfides, tungsten remobilization and synmetamorphic gold. segweb.org Google Books

3.2 Additional useful scientific papers and books

Reference	Why it is useful	Access
Cartwright & Oliver (2000), *Metamorphic fluids and their relationship to the formation of metamorphosed and metamorphogenic ore deposits*	Fluid generation, metal transport and remobilization	JCU repository record
Mishra & Upadhyay, *Hydrothermal Ore Deposits: Geochemical Attributes*	Includes a chapter on metamorphism of ore deposits	Springer book
Thomas & MacAlister (1909), *The Geology of Ore Deposits*	Historical treatment of metamorphic ore deposits, including a dedicated Chapter VII	Public-domain PDF

The 1909 book is useful for historical comparison, not as a modern scientific authority.

3.3 Which references should be used for Chapter 6?

My proposed reading priority is:

1. Vokes, Marshall and Spry – specialized scientific foundation.
2. Pohl – terminology and genetic classification.
3. Edwards and Atkinson – classical geological characteristics.
4. Robb – metamorphic fluids and hydrothermal connections.
5. Zhai, Yao and Cai – traditional classification and regional examples.
6. Recent research papers on individual deposits.

An important point about PDF availability: not every book is legally available as a free full-text PDF. The SEG publication is available as a paid PDF, Springer chapters may require institutional access, and the Cambridge frontmatter PDF is not the complete book. I have prioritized official publisher and repository links rather than directing you to unauthorized copies.

Part II. Recommended structure of the new 300+ page textbook

4. Overall textbook design

I recommend the following title:

ORE DEPOSIT GEOLOGY

Principles, Processes, Genetic Models and Global Examples

A Research-Based Textbook with Problem-Based and Project-Based Learning

For undergraduate and postgraduate students in geology, economic geology, mineral resources and related disciplines.

(6 chapters) (368 proposed pages) (156 core figures) (90 review tasks)

4.1 Proposed page allocation

Part	Pages
Preface, learning guide and introduction	10
Chapter 1. Fundamentals of Ore Deposits	46
Chapter 2. Magmatic Ore Deposits	54
Chapter 3. Hydrothermal Ore Deposits	76
Chapter 4. Weathering and Supergene Ore Deposits	42
Chapter 5. Sedimentary Ore Deposits	42
Chapter 6. Metamorphic and Metamorphosed Ore Deposits	52
Glossary	10
References	20
Answers, supplementary materials and index	16
Total	**368**

The page allocation can be adjusted after the first manuscript is written.

4.2 Recommended distribution of illustrations

Chapter	Number of figures
Chapter 1	24
Chapter 2	26
Chapter 3	40
Chapter 4	20
Chapter 5	22
Chapter 6	24
Total	**156**

I recommend a combination of:

- 40% geological conceptual diagrams and ore-forming process illustrations.
- 25% geological maps, cross-sections and global distribution maps.
- 20% photographs of deposits, outcrops, hand specimens and drill cores.
- 10% microscopic photographs and mineral assemblages.
- 5% geochemical diagrams, experimental results and quantitative models.

These proportions are design targets, not strict requirements.

A particularly important recommendation: At least half of the geological process illustrations should be original, explanatory diagrams created specifically for this textbook. Avoid relying entirely on photographs copied from other textbooks.

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5. Improved detailed table of contents and writing instructions

The following is my recommended revised six-chapter structure. It preserves your original six chapters but improves the classification, scientific depth and teaching sequence.

Chapter 1. Fundamentals of Ore Deposits and Metallogenesis

(46 pages · 24 figures)

Chapter objective

Students should understand the fundamental concepts of ore deposits, ore mineralogy, orebody geometry, mineralization processes, geological controls and genetic classification.

The chapter should establish a conceptual foundation for the remaining five chapters.

1.1. Mineral resources and the concept of ore

1.1.1. Geological resources of the Earth

Explain:

- Distribution of chemical elements in the Earth.
- Earth's crust, mantle and core as sources of mineral resources.
- Geological processes responsible for element concentration.
- Relationship between crustal evolution and mineral resources.

Students should understand that ore deposits are geological concentrations, not simply accumulations of minerals.

1.1.2. Definition and classification of ore

Explain:

- Mineral occurrence, mineralization and ore deposit.
- Ore and gangue.
- Metallic and non-metallic resources.
- Grade, tenor, tonnage and cutoff grade.
- Geological and economic definitions of ore.
- Critical and strategic mineral resources.

Emphasize that an occurrence is not automatically an ore deposit. Economic viability depends on geological, technological, environmental and market conditions.

1.1.3. Mineralogical composition of ore

Explain:

- Primary ore minerals.
- Secondary ore minerals.
- Gangue minerals.
- Accessory minerals.

- Major, minor and trace elements.
- Mineral associations and mineral paragenesis.

Include common minerals such as chalcopyrite, bornite, pyrite, sphalerite, galena, molybdenite, cassiterite, wolframite, magnetite and hematite.

1.1.4. Distribution and occurrence of valuable components

Explain:

- Independent mineral phases.
- Isomorphous substitution.
- Solid solutions.
- Microscopic inclusions.
- Adsorbed and dispersed elements.
- Distribution of valuable elements among ore minerals.

Use examples of gold in pyrite, indium in sphalerite and cobalt in nickel-bearing sulfides.

1.2. Mineralogical characteristics, textures and structures of ores

1.2.1. Ore textures and their geological significance

Discuss:

- Crystalline and granular textures.
- Disseminated textures.
- Massive textures.
- Banded textures.
- Vein and veinlet textures.
- Replacement textures.
- Colloform and crustiform textures.
- Breccia textures.
- Deformation and recrystallization textures.

Explain what each texture reveals about the mineralization process.

1.2.2. Ore structures and their geological significance

Discuss:

- Massive and disseminated structures.

- Banded and laminated structures.
- Stockwork structures.
- Breccia structures.
- Vein networks.
- Layered and rhythmic structures.

1.2.3. Mineral assemblages and paragenetic sequences

Explain how to reconstruct the sequence of mineral formation using cross-cutting relationships, mineral replacement, zoning and mineral chemistry.

1.2.4. Primary and secondary ore textures

Compare textures formed during original mineralization with those formed during oxidation, supergene enrichment, metamorphism and hydrothermal overprinting.

1.3. Ore bodies: geometry, morphology and geological controls

1.3.1. Definition and classification of ore bodies

Discuss the difference between an ore deposit and an individual orebody.

1.3.2. Morphology of ore bodies

Explain:

- Tabular bodies.
- Lenticular bodies.
- Stratiform bodies.
- Stratabound bodies.
- Vein-type bodies.
- Stockwork bodies.
- Irregular and massive bodies.
- Pipe and chimney-shaped bodies.
- Disseminated mineralization.

1.3.3. Attitude and spatial relationships of ore bodies

Discuss strike, dip, plunge, thickness, continuity and spatial relationships with host rocks.

1.3.4. Geological controls on orebody geometry

Explain structural, stratigraphic, lithological, chemical and permeability controls.

1.4. Ore deposits and genetic classification

1.4.1. Historical development of ore deposit classification

Compare historical classifications with modern genetic and process-based classifications.

1.4.2. Classification according to geological processes

Explain:

- Magmatic processes.
- Magmatic-hydrothermal processes.
- Hydrothermal processes.
- Sedimentary processes.
- Diagenetic processes.
- Weathering and supergene processes.
- Metamorphic and metamorphogenic processes.

1.4.3. Ore-forming systems

Introduce the integrated ore-forming system:

The ore-forming system

1. Source

Metals, sulfur, heat and fluids

2. Transport

Magma, fluids, groundwater and sediment transport

3. Deposition and concentration

Precipitation, crystallization, replacement and mechanical concentration

4. Preservation

Burial, tectonic stability and geological preservation

1.4.4. Multiple-stage mineralization and overprinting

Explain how different geological events may produce composite deposits.

1.5. Geochemical and geotectonic foundations of ore formation

1.5.1. Geochemical behavior of elements

Explain:

- Lithophile elements.
- Siderophile elements.
- Chalcophile elements.
- Atmophile elements.
- Compatible and incompatible elements.

1.5.2. Element partitioning and concentration

Discuss partition coefficients, crystal-melt partitioning, mineral-fluid partitioning and mass balance.

1.5.3. Physicochemical conditions of mineralization

Introduce:

- Temperature.
- Pressure.
- Oxygen fugacity.
- Sulfur fugacity.
- pH and Eh.
- Solubility.
- Metal complexing.
- Mineral saturation.

1.5.4. Tectonic setting and metallogenic provinces

Explain relationships between mineral deposits and:

- Divergent plate boundaries.
- Convergent plate boundaries.
- Island arcs.
- Continental arcs.
- Continental rifts.
- Collisional orogens.
- Stable cratons.
- Large igneous provinces.

1.6. Methods for investigating ore genesis

1.6.1. Geological and petrographic observations

Field relationships, ore microscopy, thin-section examination and mineral identification.

1.6.2. Mineral chemistry and whole-rock geochemistry

XRF, XRD, EPMA, LA-ICP-MS and ICP-MS, including what each technique can reveal.

1.6.3. Isotopes, geochronology and fluid inclusions

Explain how geologists determine the sources and timing of mineralization.

1.6.4. Construction and testing of genetic models

Teach students to combine geological, mineralogical, geochemical and geochronological evidence, recognize uncertainty and compare alternative hypotheses.

Chapter 2. Magmatic Ore Deposits

(54 pages · 26 figures)

Chapter objective

Students should understand how magma generation, differentiation, element partitioning and liquid immiscibility produce economically important concentrations of metals.

2.1. Formation and evolution of magma

2.1.1. Origin and sources of magma

Discuss mantle melting, crustal melting, mantle metasomatism and the influence of tectonic settings.

2.1.2. Partial melting and magma generation

Explain:

- Degree of partial melting.
- Pressure and temperature.
- Water content.
- Source-rock composition.
- Trace-element enrichment.

2.1.3. Magma differentiation and evolution

Discuss fractional crystallization, assimilation, magma mixing, crystal accumulation and recharge.

2.1.4. Magmatic volatiles and magma fertility

Explain how H_2O, CO_2, sulfur, chlorine, fluorine and other volatile components influence metal enrichment and ore-forming potential.

2.1.5. Geochemical evolution of magmatic systems

Explain how elements are distributed among melt, crystals, sulfide liquid and oxide phases.

2.2. Ore-forming mechanisms in magmatic systems

2.2.1. Fractional crystallization and crystal accumulation

Explain how mineral phases crystallize and become concentrated within magma chambers.

2.2.2. Liquid immiscibility

Discuss:

- Silicate-sulfide immiscibility.
- Silicate-oxide immiscibility.
- Formation and separation of immiscible liquids.

2.2.3. Sulfide saturation and metal concentration

Explain the role of sulfur availability, magma contamination, magma mixing and fractional crystallization.

2.2.4. Sulfide-silicate partition coefficients and the R-factor

Explain why Ni, Cu, Co and PGE become concentrated in sulfide liquids.

2.2.5. Filter pressing and magma transport

Explain how compaction, crystal accumulation and liquid migration influence mineral concentration.

2.3. Main types of mafic-ultramafic magmatic deposits

2.3.1. Chromite deposits

Discuss stratiform and podiform chromite deposits, their tectonic environments and formation mechanisms.

Case studies: Bushveld Complex and ophiolite-hosted chromite.

2.3.2. Magmatic Ni-Cu-Co sulfide deposits

Discuss komatiite-hosted, conduit-hosted and intrusion-related sulfide deposits.

Case studies: Norilsk-Talnakh, Sudbury and Kambalda.

2.3.3. Platinum-group element deposits

Explain PGE concentration in layered intrusions, reef-type mineralization, sulfide saturation and competing genetic models.

Case studies: Merensky Reef and Great Dyke.

2.3.4. Fe-Ti-V oxide deposits

Explain the concentration of magnetite, ilmenite and vanadium-bearing oxides.

Discuss anorthosite-associated and layered intrusion-related examples.

2.4. Main types of felsic and alkaline magmatic deposits

2.4.1. Pegmatite deposits

Explain extreme magma fractionation, incompatible elements and enrichment in Li, Cs, Ta, Nb, Be and Sn.

2.4.2. Granite-related Sn-W-Mo deposits

Discuss evolved granitic magma, volatile-rich residual melts and the transition from magmatic to hydrothermal mineralization.

2.4.3. Carbonatite deposits

Explain carbonatitic magma generation and enrichment of rare-earth elements, Nb and phosphate.

Case studies: Bayan Obo and Mountain Pass, with discussion of their genetic interpretations.

2.4.4. Kimberlite and lamproite diamond deposits

Explain mantle sources, diamond stability, magma ascent and preservation of diamonds.

2.4.5. Alkaline and peralkaline magmatic deposits

Discuss rare metals and REE mineralization associated with alkaline and peralkaline igneous systems.

2.4.6. Comparative genetic models

Compare the formation of deposits in mafic, ultramafic, felsic, alkaline and carbonatitic magmatic systems.

2.4.7. Integrated magmatic metallogenesis

Explain why different magmatic processes may generate multiple deposit types within the same magmatic province.

Chapter 3. Hydrothermal Ore Deposits

(76 pages · 40 figures)

Chapter objective

Students should understand hydrothermal fluid generation, migration, metal transport, fluid-rock reactions and the physical and chemical mechanisms of mineral precipitation.

This should be the most scientifically detailed chapter because hydrothermal systems account for a wide variety of economically important deposit types.

3.1. Hydrothermal systems and ore-forming fluids

3.1.1. Concept and characteristics of hydrothermal systems

Explain the components of a hydrothermal system and its thermal, chemical and hydrological evolution.

3.1.2. Sources of hydrothermal fluids

Discuss:

- Magmatic fluids.
- Meteoric water.
- Seawater.
- Basinal fluids.
- Metamorphic fluids.
- Mixed-source fluids.

3.1.3. Chemical composition and properties of hydrothermal fluids

Explain fluid salinity, density, gas composition, pH, Eh and major dissolved components.

3.1.4. Metal transport and complex formation

Discuss chloride, bisulfide, hydroxide and other important metal complexes.

3.1.5. Fluid migration and permeability

Explain the role of fractures, faults, shear zones, porous rocks, fluid pressure and hydrofracturing.

3.1.6. Mineral precipitation mechanisms

Discuss:

- Cooling.
- Pressure reduction.
- Boiling.
- Fluid mixing.
- Fluid-rock reaction.
- Changes in pH and redox conditions.
- Sulfidation and oxidation.

3.1.7. Hydrothermal alteration and mineral zoning

Explain potassic, phyllic, propylitic, argillic, silicic, advanced argillic, greisen and other alteration assemblages.

3.2. Magmatic-hydrothermal ore deposits

3.2.1. Porphyry Cu-Mo-Au deposits

Explain magma evolution, fluid exsolution, stockwork formation, alteration zoning and metal precipitation.

Case studies: Escondida, Chuquicamata, Grasberg and Bingham Canyon.

3.2.2. Epithermal Au-Ag deposits

Discuss:

- Low-sulfidation.
- Intermediate-sulfidation.
- High-sulfidation systems.

Explain boiling, fluid mixing, wall-rock reactions and precious-metal deposition.

3.2.3. Skarn deposits

Explain contact metamorphism, metasomatism, prograde and retrograde alteration, and the formation of garnet-pyroxene assemblages.

Discuss Fe, Cu, Au, W, Sn, Zn and Pb skarn deposits.

3.2.4. Greisen deposits

Explain granite-related hydrothermal alteration, greisenization, quartz-muscovite assemblages and Sn-W mineralization.

3.2.5. Albitite-related mineralization

Discuss sodium metasomatism, fluid-rock reaction and relevant uranium and rare-metal systems. Explain that albitite is an alteration/host-rock description rather than a single universally defined genetic deposit class.

3.2.6. Carbonate-replacement and polymetallic replacement deposits

Explain replacement mechanisms in carbonate rocks, fluid pathways and formation of sulfide-rich bodies.

3.2.7. Intrusion-related gold and polymetallic vein deposits

Discuss intrusion-related fluids, vein systems, mineral zoning and structural controls.

3.2.8. Iron oxide-copper-gold deposits

Explain the characteristics and competing genetic interpretations of IOCG systems.

3.2.9. Iron oxide-apatite deposits

Discuss magnetite-apatite mineralization and its relationships with magmatic and hydrothermal processes.

3.2.10. Hydrothermal systems associated with alkaline magmatism

Explain hydrothermal REE, Nb and other critical-metal systems.

3.2.11. Comparative characteristics of magmatic-hydrothermal deposits

Compare temperature, depth, fluid composition, mineral assemblages and alteration.

3.2.12. Integrated magmatic-hydrothermal evolution

Explain how a single intrusive complex can generate different mineralization styles at different depths and stages.

3.3. Volcanic and seafloor hydrothermal deposits

3.3.1. Volcanogenic massive sulfide deposits

Discuss VMS formation, submarine volcanic environments, hydrothermal convection and sulfide precipitation.

3.3.2. Major VMS subtypes

Compare Kuroko, Cyprus, Besshi and Noranda-type examples.

3.3.3. Modern seafloor hydrothermal systems

Explain black smokers, white smokers, hydrothermal chimneys and modern metal precipitation.

3.3.4. Geological and geochemical characteristics of VMS deposits

Discuss mineral zoning, alteration pipes, massive sulfides and footwall stockworks.

3.3.5. Relationship between modern hydrothermal systems and ancient VMS deposits

Explain how modern seafloor observations are used to reconstruct ancient mineralization.

3.4. Orogenic and other metamorphic-fluid-related hydrothermal deposits

3.4.1. Orogenic gold deposits

Explain their tectonic setting, structural controls, quartz-carbonate veins, mineral assemblages and fluid characteristics.

3.4.2. Origin and evolution of metamorphic fluids

Discuss devolatilization, fluid flow during deformation and the possible sources of metals.

3.4.3. Gold transport and precipitation

Explain gold-bisulfide complexing, sulfidation, cooling, pressure changes and fluid-rock reaction.

3.4.4. Carlin-type gold deposits

Discuss their geological characteristics, invisible gold, arsenian pyrite, alteration and competing fluid-source models.

3.4.5. Comparison of orogenic and Carlin-type gold systems

Explain similarities and differences in fluid sources, host rocks, structures and mineralization.

3.5. Hydrothermal deposits associated with sedimentary basins

3.5.1. Basinal hydrothermal fluids

Discuss compaction, evaporite dissolution, fluid migration, basin architecture and metal sources.

3.5.2. Mississippi Valley-type deposits

Explain carbonate-hosted Pb-Zn mineralization, basinal brines, fluid mixing and sulfide precipitation.

3.5.3. Sedimentary exhalative deposits

Explain SEDEX formation, basin brines, subseafloor fluid migration, seafloor discharge and sulfide accumulation.

3.5.4. Sediment-hosted stratiform copper deposits

Discuss red-bed copper, Kupferschiefer and Central African Copperbelt examples.

3.5.5. Sandstone-hosted uranium deposits

Explain oxidized uranium-bearing groundwater, reducing environments and roll-front mineralization.

3.5.6. Unconformity-related uranium deposits

Discuss basin-basement relationships, fluid pathways, redox reactions and uranium precipitation.

3.5.7. Comparative models of basin-related hydrothermal deposits

Compare MVT, SEDEX, sediment-hosted Cu and uranium deposits.

3.6. Integrated interpretation of hydrothermal systems

3.6.1. Reconstruction of fluid evolution

Combine fluid inclusions, isotopes, mineral chemistry and alteration.

3.6.2. Recognition of multiple mineralization stages

Teach students to distinguish primary mineralization from later overprinting.

3.6.3. Evaluation of competing genetic models

Teach scientific reasoning through evidence, uncertainty and alternative hypotheses.

3.6.4. Integrated hydrothermal case study

Students reconstruct an entire hydrothermal system from geological maps, mineral assemblages, fluid inclusion data and geochemical evidence.

Chapter 4. Ore Deposits Formed by Weathering and Supergene Processes

(42 pages · 20 figures)

Chapter objective

Students should understand how surface weathering, groundwater circulation, oxidation, leaching and secondary enrichment generate economically important mineral concentrations.

Important structural improvement: Your original Chapter 4 overlaps with Chapter 5 because both address sedimentary environments. I recommend changing Chapter 4 to focus specifically on weathering, residual concentration, infiltration and supergene enrichment.

4.1. Geological and geochemical conditions of weathering-related deposits

4.1.1. Weathering processes and regolith formation

Explain physical, chemical and biological weathering.

4.1.2. Factors controlling weathering and mineral concentration

Discuss climate, parent-rock composition, topography, drainage, time and tectonic stability.

4.1.3. Geochemical mobility during weathering

Explain dissolution, hydrolysis, oxidation, leaching, adsorption and precipitation.

4.1.4. Weathering profiles and vertical mineral zoning

Discuss soil, laterite, ferricrete, saprolite, weathered bedrock and groundwater zones.

4.2. Residual ore deposits

4.2.1. Principles of residual concentration

Explain the removal of mobile components and residual enrichment of relatively immobile elements.

4.2.2. Bauxite deposits

Discuss the formation of aluminum-rich residual profiles.

4.2.3. Lateritic nickel deposits

Explain Ni-Co enrichment in weathered ultramafic rocks.

4.2.4. Lateritic iron and manganese deposits

Discuss the concentration of Fe and Mn through weathering and precipitation.

4.2.5. Ion-adsorption rare-earth element deposits

Explain the weathering of REE-rich rocks and adsorption of REE onto clay minerals.

4.2.6. Comparative characteristics of residual deposits

Compare bauxite, lateritic Ni and ion-adsorption REE deposits.

4.3. Supergene alteration and enrichment

4.3.1. Oxidation of primary sulfide deposits

Explain the interaction between oxygenated groundwater and sulfide minerals.

4.3.2. Formation of gossans and oxidation zones

Discuss iron oxides, hydroxides and secondary minerals.

4.3.3. Secondary sulfide enrichment

Explain downward migration of metals and their reprecipitation below the oxidation zone.

4.3.4. Supergene enrichment of copper deposits

Discuss chalcocite, covellite and secondary copper mineralization.

4.3.5. Supergene enrichment of other metals

Compare relevant gold, silver and other sulfide-related systems.

4.4. Infiltration and groundwater-related deposits

4.4.1. Groundwater migration and mineral precipitation

Discuss groundwater flow, permeability, redox interfaces and geochemical traps.

4.4.2. Roll-front uranium deposits

Explain uranium transport and reduction in permeable sedimentary formations.

4.4.3. Calcrete-hosted uranium deposits

Discuss evaporation, carbonate-rich groundwater and uranium precipitation.

4.5. Evolution and preservation of weathering-related deposits

4.5.1. Geological preservation of weathering profiles

Discuss erosion, burial, uplift and climate evolution.

4.5.2. Comparative case studies

Use examples from tropical laterites, bauxite plateaus and supergene copper districts.

Chapter 5. Sedimentary Ore Deposits

(42 pages · 22 figures)

Chapter objective

Students should understand how mechanical transport, sedimentation, chemical precipitation and biological activity concentrate valuable minerals.

5.1. Fundamental conditions and processes of sedimentary mineralization

5.1.1. Principles of sedimentary ore formation

Explain sediment transport, deposition, sorting, chemical precipitation and biological processes.

5.1.2. Physical conditions of sediment concentration

Discuss grain size, density, shape, flow velocity, settling velocity and hydraulic sorting.

5.1.3. Chemical conditions of sedimentary mineralization

Explain pH, Eh, salinity, redox stratification and water chemistry.

5.1.4. Sedimentary basin evolution and ore formation

Discuss basin development, sedimentary facies, water circulation and preservation.

5.2. Placer deposits

5.2.1. Formation mechanisms of placer deposits

Explain hydraulic sorting, density differences and sedimentary concentration.

5.2.2. Fluvial placer deposits

Discuss gold, cassiterite, diamond and other heavy-mineral concentrations in river systems.

5.2.3. Beach and coastal placer deposits

Explain wave action, longshore currents and heavy-mineral accumulation.

5.2.4. Ancient and paleoplacer deposits

Discuss conglomerate-hosted gold and uranium deposits, including the debated origin of Witwatersrand mineralization.

5.2.5. Comparative characteristics of placer deposits

Compare modern and ancient placers and their geological preservation.

5.3. Chemical sedimentary deposits

5.3.1. Banded iron formations

Explain iron precipitation, ocean chemistry, hydrothermal input, microbial processes and the evolution of atmospheric oxygen.

5.3.2. Oolitic iron deposits

Discuss iron-rich sedimentation, chemical conditions and sedimentary structures.

5.3.3. Sedimentary manganese deposits

Explain Mn concentration in marine basins and chemical precipitation.

5.3.4. Marine manganese nodules and crusts

Discuss ocean-floor mineralization and the role of slow precipitation.

5.3.5. Evaporite deposits

Explain evaporation, brine concentration and sequential mineral precipitation.

5.3.6. Phosphorite deposits

Discuss marine productivity, upwelling and phosphorus cycling.

5.3.7. Comparative chemical sedimentary systems

Compare iron, manganese, phosphate and evaporite deposits.

5.4. Biochemical and organic-associated mineral deposits

5.4.1. Role of microorganisms in mineral concentration

Explain microbial mediation of oxidation-reduction reactions and mineral precipitation.

5.4.2. Biochemical iron and manganese mineralization

Discuss microbial contributions to Fe and Mn cycling.

5.4.3. Organic-rich sediment-hosted metal concentrations

Discuss the association of organic matter with uranium, vanadium, nickel and molybdenum in appropriate sedimentary environments.

5.5. Sedimentary metallogeny and global distribution

5.5.1. Evolution of sedimentary mineralization through geological time

Explain the relationship between ocean chemistry, atmospheric oxygen and sedimentary mineral deposits.

5.5.2. Global distribution of sedimentary deposits

Discuss major metallogenic regions and sedimentary basins.

5.5.3. Integrated sedimentary basin case studies

Students reconstruct the depositional environment and concentration mechanisms of an actual deposit.

Chapter 6. Metamorphic and Metamorphosed Ore Deposits

(52 pages · 24 figures)

Chapter objective

Students should understand how metamorphism creates new mineral concentrations, modifies pre-existing deposits and mobilizes metals through metamorphic fluids.

This chapter should be written as a scientifically independent chapter, but with strong connections to Chapters 1 and 3.

6.1. Fundamental concepts of metamorphic ore deposits

6.1.1. Definition and classification of metamorphic ore deposits

Introduce the three concepts:

- Metamorphic deposits.
- Metamorphosed deposits.
- Metamorphogenic deposits.

6.1.2. Geological conditions of metamorphism

Discuss:

- Temperature.
- Pressure.
- Fluid composition.
- Deformation.
- Metamorphic grade.
- Metamorphic facies.

6.1.3. Types of metamorphic environments

Explain regional, contact, dynamic, burial, subduction-related and high-grade metamorphism.

6.2. Metamorphic processes and metal mobilization

6.2.1. Metamorphic reactions and mineral transformation

Explain recrystallization, neocrystallization, mineral reactions and metamorphic differentiation.

6.2.2. Generation of metamorphic fluids

Discuss dehydration, decarbonation and devolatilization reactions.

6.2.3. Fluid migration and metal transport

Explain fluid pressure, structural permeability, shear zones, fluid-rock interaction and metal complexing.

6.2.4. Ore-forming mechanisms during metamorphism

Discuss cooling, pressure reduction, sulfidation, redox changes, fluid mixing and mineral precipitation.

6.3. Metamorphosed pre-existing ore deposits

6.3.1. Metamorphosed VMS and SEDEX deposits

Explain changes in mineralogy, texture, structure and geometry.

6.3.2. Metamorphosed Ni-Cu-PGE deposits

Discuss the effects of deformation and metamorphism on sulfide mineral assemblages.

6.3.3. Metamorphosed iron formations

Explain recrystallization, metamorphic differentiation and formation of high-grade iron ores.

6.3.4. Metamorphic remobilization of metals

Discuss metal redistribution, vein formation, local upgrading and downgrading.

6.3.5. Geological criteria for identifying metamorphosed deposits

Teach students to distinguish original mineralization from metamorphic modification using textures, mineral chemistry, geochronology and structural relationships.

6.4. Metamorphogenic ore deposits

6.4.1. Orogenic gold deposits and metamorphic fluids

Discuss the relationship between regional metamorphism, deformation and gold mineralization.

This section should cross-reference Chapter 3.4, rather than repeat its entire deposit description.

6.4.2. Metamorphic tungsten mineralization

Use the Felbertal scheelite deposit as a case study of metamorphic remobilization and tungsten mineralization.

6.4.3. Metamorphic gold and base-metal deposits in high-grade terranes

Discuss mineralization in amphibolite- and granulite-facies environments.

6.4.4. Comparison of metamorphic and metamorphogenic models

Teach students how to evaluate whether metamorphism caused new mineralization or merely redistributed an older orebody.

6.5. Mineral deposits formed through metamorphic mineral reactions

6.5.1. Graphite deposits

Explain the transformation of carbonaceous material into crystalline graphite during metamorphism.

6.5.2. Kyanite, sillimanite and andalusite deposits

Discuss Al-silicate mineral formation and the influence of pressure-temperature conditions.

6.5.3. Talc, magnesite and other metamorphic industrial minerals

Explain fluid-rock reactions, metamorphic alteration of ultramafic rocks and mineral assemblage formation.

6.5.4. Contact metamorphic industrial mineral deposits

Discuss wollastonite, garnet and related mineralization, while distinguishing isochemical metamorphic reactions from metasomatic skarn formation.

6.6. Global examples and genetic interpretation

6.6.1. Broken Hill Pb-Zn-Ag deposit

A case study of a complex, highly metamorphosed mineral system and competing interpretations of its original formation.

6.6.2. Kambalda Ni sulfide deposits

Discuss original komatiite-hosted mineralization and later metamorphic modification.

6.6.3. Felbertal tungsten deposit

Examine tungsten mineralization and metamorphic remobilization.

6.6.4. Yilgarn Craton gold deposits

Discuss gold mineralization in metamorphic terranes and evidence for synmetamorphic processes.

These case studies should emphasize that a deposit's present mineralogy and geometry may not directly represent its original form.

Part III. How to write the textbook using PBL teaching theory

6. Recommended educational philosophy

I recommend combining six teaching approaches:

Teaching theory	Application in the textbook
Constructivist learning	Students construct geological explanations using previous knowledge and new evidence
Problem-based learning	Students solve geological problems with incomplete information
Project-based learning	Students complete longer research projects and develop geological models
Inquiry-based learning	Students formulate questions, investigate evidence and test hypotheses
Case-based learning	Students study actual mineral deposits and their geological histories
Collaborative learning	Students discuss competing interpretations and defend their conclusions

This recommendation is supported by research in geoscience education.

For example:

- Smith, Hoersch and Gordon (1995) discussed problem-based learning in undergraduate geology, including mineralogy and metamorphic petrology. Taylor & Francis Online
- *Geology and Environment: A Problem-Based Learning Study in Higher Education* (2021) investigated PBL in university geology education. DOI
- Holley (2017) demonstrated the use of geological case studies and active learning with mining engineering students. ERIC
- The National Association of Geoscience Teachers presented a particularly useful economic geology teaching case involving the Pebble deposit in Alaska. NAGT

6.1. Distinguish problem-based learning from project-based learning

Although both are commonly abbreviated PBL, they should have different roles in your textbook.

Problem-based learning

(Shorter learning cycle)

Students receive a geological problem and investigate it using scientific evidence.

Example:

A geological core contains disseminated chalcopyrite, bornite, quartz veinlets and potassic alteration. What type of mineral deposit might this represent? What additional evidence is necessary to confirm your interpretation?

Expected result: A reasoned hypothesis and supporting evidence.

Project-based learning

(Longer research cycle)

Students complete an integrated investigation and produce a substantial research output.

Example:

Reconstruct the geological evolution of a major porphyry copper deposit. Investigate its tectonic setting, magmatic evolution, hydrothermal alteration, mineralization stages and genetic model.

Expected result: A research report, geological model, annotated cross-section and oral presentation.

6.2. Standard structure for every textbook subsection

This is probably the most important practical recommendation for writing the new textbook.

Every subsection should follow a consistent teaching structure.

Component	What the author should write
1. Geological problem	Introduce an interesting geological situation
2. Learning objectives	State what students should understand
3. Fundamental concepts	Explain the scientific principles
4. Geological evidence	Present maps, photographs, mineral assemblages and datasets
5. Ore-forming mechanism	Explain the physical and chemical processes
6. Real-world example	Introduce an actual mineral deposit
7. Thinking problem	Ask students to interpret or evaluate evidence
8. Model answer	Explain the scientific reasoning
9. PBL activity	Provide a small investigation or group assignment
10. Key knowledge	Summarize the essential conclusions

For example, when writing about porphyry copper deposits, do not begin with a long description of their characteristics.

Instead:

Opening problem

A large intrusive complex is exposed in a continental volcanic arc. Copper mineralization occurs as disseminated sulfides and quartz stockwork veins. Several alteration zones surround the intrusive rocks.

Why does copper concentrate in this particular geological environment?

Then explain:

1. The geological setting.
2. The composition and evolution of magma.
3. The formation of magmatic fluids.
4. Fluid migration and pressure changes.

5. Metal transport and precipitation.
6. Hydrothermal alteration.
7. The formation of the orebody.
8. Evidence supporting the genetic model.

This structure makes the scientific process much clearer to students.

6.3. How to use thinking problems

Thinking problems should not merely ask students to repeat definitions.

They should require students to:

- Explain geological mechanisms.
- Compare two competing models.
- Interpret geological observations.
- Predict mineral assemblages.
- Analyze geological cross-sections.
- Evaluate geochemical evidence.
- Identify missing evidence.
- Defend a scientific interpretation.

I recommend placing one or two thinking problems after every major subsection, followed by a short model answer.

For research-based questions, the answer should explain the reasoning rather than claim that only one interpretation is always correct.

Part IV. Thinking problems and model answers

The following is a proposed question bank. These questions can be inserted into the appropriate sections of the textbook.

Chapter 1. Fundamentals of Ore Deposits

Section	Thinking problem	Model answer
1.1	Why is every mineral occurrence not necessarily an ore deposit?	Because an ore deposit requires a sufficiently concentrated and recoverable resource under relevant economic, technological and environmental conditions.
1.1	Can the same mineral deposit be an ore deposit today but not in the future?	Yes. Changes in commodity prices, technology, regulations and environmental constraints can change its economic status.
1.2	Why can two ores with similar mineral composition have different textures?	They may have formed under different crystallization, replacement, deformation or recrystallization conditions.
1.2	What can a mineral replacing another mineral tell us?	It indicates a later reaction or alteration event and helps reconstruct the paragenetic sequence.
1.3	Why do some ore deposits form veins while others form stratiform bodies?	Their geometries reflect differences in geological structures, host-rock properties, depositional processes and mineralization mechanisms.
1.3	How can folding change orebody geometry without changing the original metal content?	Folding can deform and redistribute the spatial shape of the body without necessarily changing its total metal inventory.
1.4	Why should descriptive and genetic classifications be used together?	Descriptive classification identifies observable characteristics; genetic classification explains their origin.
1.4	Can one deposit belong to more than one genetic category?	Yes. Deposits can undergo multiple mineralization and modification events.

Section	Thinking problem	Model answer
1.5	Why do some elements concentrate in sulfide minerals rather than silicate minerals?	Their geochemical affinities and partition coefficients favor sulfide phases under appropriate conditions.
1.5	How can changing oxygen fugacity influence ore formation?	It changes element valence, mineral stability and the solubility of metals in melts and fluids.
1.6	Why are isotopic studies useful in ore genesis?	Isotopes can constrain sources of fluids, sulfur and metals, as well as geological timing, depending on the isotope system.
1.6	Why is one analytical method rarely sufficient to establish a deposit's origin?	Different datasets constrain different aspects of the system; multiple independent lines of evidence are needed.

Chapter 2. Magmatic Ore Deposits

Section	Thinking problem	Model answer
2.1	Why are some magmas more favorable for ore formation than others?	They may inherit metals from enriched sources and have suitable composition, volatile contents, redox conditions and differentiation histories.
2.1	How can partial melting influence the concentration of incompatible elements?	Small degrees of partial melting can produce melts enriched in elements that preferentially enter the melt rather than remain in the residue.
2.2	Why does sulfide liquid efficiently concentrate Ni, Cu and PGE?	These elements preferentially partition into sulfide liquid relative to silicate melt.

Section	Thinking problem	Model answer
2.2	What happens if external sulfur is added to a magma?	It may trigger sulfide saturation and initiate segregation of a metal-rich sulfide liquid.
2.3	Why do chromite deposits commonly occur in layered mafic-ultramafic intrusions?	Changes in magma composition, phase equilibria, replenishment and crystallization can produce chromite saturation and accumulation.
2.3	Why is the R-factor important in Ni-Cu-PGE mineralization?	It describes the mass of silicate magma that interacts with sulfide liquid and helps explain the degree of metal enrichment.
2.4	Why can highly evolved granitic magmas produce rare-metal pegmatites?	Incompatible elements and volatiles become concentrated in residual melts during prolonged differentiation.
2.4	What is the principal difference between carbonatite and ordinary granitic magma systems?	They differ in melt composition, petrogenesis, element partitioning and the types of mineralization they can generate.

Chapter 3. Hydrothermal Ore Deposits

Section	Thinking problem	Model answer
3.1	How can geologists distinguish magmatic fluids from meteoric or basinal fluids?	By integrating stable isotopes, fluid inclusion properties, salinity, gas composition, geological relationships and fluid-rock interaction evidence.
3.1	Why does a metal dissolved in a	Changes in temperature, pressure, ligand availability, pH, redox conditions or fluid composition can reduce its solubility.

Section	Thinking problem	Model answer
	hydrothermal fluid eventually precipitate?	
3.2	Why are porphyry copper deposits commonly associated with large intrusive complexes?	Large, water- and sulfur-rich magmatic systems can release substantial quantities of metal-bearing fluids and create extensive hydrothermal systems.
3.2	Why do porphyry systems commonly have alteration zoning?	Fluid composition, temperature, pressure and fluid-rock reactions vary spatially and through time.
3.2	What distinguishes a skarn from a typical porphyry copper deposit?	Skarns are characterized by metasomatic replacement, commonly of carbonate rocks, and calc-silicate mineral assemblages.
3.3	How do modern black smokers help explain ancient VMS deposits?	They provide direct observations of submarine hydrothermal circulation, metal transport and sulfide precipitation.
3.3	Why do VMS deposits commonly occur near volcanic centers?	Volcanic heat and permeability can drive seawater circulation and hydrothermal mineralization.
3.4	Why are quartz-carbonate veins common in many orogenic gold deposits?	They reflect the movement and precipitation of aqueous-carbonic fluids in structurally controlled pathways.
3.4	Is the presence of CO_2-rich fluid inclusions sufficient to prove metamorphic fluid origin?	No. Similar fluid compositions can have multiple origins; isotopic, geochronological and geological evidence is also required.

Section	Thinking problem	Model answer
3.5	What is the fundamental difference between MVT and SEDEX deposits?	MVT deposits are typically carbonate-hosted replacement systems formed from basinal fluids, whereas SEDEX deposits are commonly stratiform sulfide accumulations associated with basin-floor fluid discharge.
3.5	Why are redox boundaries important in sandstone-hosted uranium deposits?	Uranium can be transported in oxidized groundwater and precipitated where reducing conditions convert it into less soluble forms.
3.6	How can two competing genetic models be evaluated?	Compare their predictions with field relationships, mineral textures, geochemistry, isotopes, fluid inclusions and age constraints.

Chapter 4. Weathering and Supergene Ore Deposits

Section	Thinking problem	Model answer
4.1	Why are thick lateritic profiles common in some tropical regions?	Warm temperatures, high rainfall, suitable drainage and long periods of weathering can promote intense chemical alteration.
4.1	Can the same parent rock produce different weathering-related deposits in different climates?	Yes. Climate, topography, drainage and time influence mineral stability and element mobility.
4.2	What is the difference between residual enrichment and secondary enrichment?	Residual enrichment concentrates relatively immobile elements as mobile components are removed; secondary enrichment involves transport and redeposition of elements.

Section	Thinking problem	Model answer
4.2	Why can ultramafic rocks be favorable for lateritic nickel deposits?	They commonly contain Ni-bearing minerals that release nickel during intense chemical weathering.
4.3	Why can copper concentration increase below an oxidized sulfide zone?	Copper released during oxidation can migrate downward and precipitate as secondary sulfides in a suitable reducing environment.
4.3	Why is a gossan not necessarily evidence of an economically valuable sulfide deposit?	It indicates oxidation of iron-bearing minerals but does not establish the presence or grade of valuable metals.
4.4	What geological conditions favor roll-front uranium mineralization?	Permeable sediments, oxidized uranium-bearing groundwater and a suitable reducing boundary.
4.5	Why are ancient weathering profiles important?	They record past climate, landscape evolution, weathering conditions and potentially preserved mineral enrichment.

Chapter 5. Sedimentary Ore Deposits

Section	Thinking problem	Model answer
5.1	Why do heavy minerals become concentrated in some sedimentary environments?	Differences in density, grain size, shape and hydrodynamic behavior produce selective sediment sorting.
5.1	Why is the depositional environment important in	It controls sediment supply, energy conditions, water chemistry and preservation.

Section	Thinking problem	Model answer
	predicting sedimentary mineralization?	
5.2	Why can gold accumulate in river bends and behind obstacles?	Changes in flow energy and hydraulic sorting allow dense gold particles to settle and accumulate.
5.2	Why are ancient placers sometimes different from modern placers?	They may have experienced compaction, diagenesis, deformation, metamorphism and later modification.
5.3	Why are BIF deposits important in understanding Earth's early history?	Their formation is connected to ancient ocean chemistry, iron cycling and the evolution of atmospheric and oceanic oxygenation.
5.3	What conditions promote phosphorite formation?	High biological productivity, phosphorus cycling, sedimentary reworking and appropriate chemical conditions.
5.4	How can microorganisms influence metal concentration?	They can modify redox conditions, produce reactive compounds and promote mineral precipitation or dissolution.
5.5	Why do sedimentary ore deposits show different distributions through geological time?	Changes in tectonics, ocean chemistry, atmospheric composition, biological activity and sedimentary environments affect mineralization.

Chapter 6. Metamorphic and Metamorphosed Ore Deposits

Section	Thinking problem	Model answer
6.1	What is the difference between metamorphic and	Metamorphic deposits form through metamorphic processes, whereas metamorphosed deposits existed

Section	Thinking problem	Model answer
	metamorphosed deposits?	before metamorphism and were subsequently modified.
6.1	Why should metamorphogenic deposits be distinguished from metamorphosed deposits?	Metamorphogenic deposits involve the formation or substantial upgrading of mineralization through metamorphic metal transport and deposition.
6.2	How can dehydration reactions contribute to ore formation?	They release fluids that may transport metals and promote mineral precipitation in favorable structural and chemical traps.
6.2	Why are shear zones important in metamorphic mineralization?	They can focus fluid movement and provide pathways and sites for fluid-rock reactions.
6.3	How can geologists determine whether sulfide mineralization predates metamorphism?	They examine preserved depositional textures, cross-cutting relationships, mineral generations and independent age constraints.
6.3	Does recrystallization necessarily increase ore grade?	No. It can concentrate, redistribute or dilute metals depending on mineral reactions and mass transfer.
6.4	How can metamorphic fluids transport gold?	Gold may be transported in aqueous complexes, particularly sulfur-bearing complexes, depending on fluid chemistry.
6.4	Why is the origin of orogenic gold still scientifically debated?	Different deposits may involve varying fluid sources, metal sources and timing relationships, and the available evidence is not always unique.
6.5	How does metamorphic grade influence Al-silicate mineral assemblages?	Pressure-temperature conditions determine the stability fields of andalusite, kyanite and sillimanite.

Section	Thinking problem	Model answer
6.6	Why is Broken Hill a valuable teaching case?	It illustrates the complexity of highly metamorphosed ore systems and the challenge of distinguishing original mineralization from later modification.

Part V. Fifteen review tasks at the end of each chapter

I recommend making these tasks more substantial than ordinary examination questions.

Each chapter should include 15 tasks distributed among five categories:

- Conceptual understanding.
- Geological data interpretation.
- Genetic model construction.
- Real deposit investigation.
- Independent research and scientific communication.

The following assignments can be used for individual work, small-group work or semester assessments.

Chapter 1: Fundamentals of Ore Deposits

No.	Review task	Expected output
1	Explain the differences between mineral occurrence, mineralization, mineral resources and ore reserves.	Comparative scientific report
2	Select ten important ore minerals and explain their chemical composition and economic significance.	Mineralogical database
3	Examine five ore specimens and classify their textures and structures.	Illustrated petrographic report

No.	Review task	Expected output
4	Explain how mineral paragenesis can be reconstructed from cross-cutting relationships.	Paragenetic diagram
5	Compare tabular, lenticular, stratiform, vein and stockwork ore bodies.	Geological drawings
6	Construct a geological cross-section showing the relationship between host rocks, structures and ore bodies.	Annotated cross-section
7	Explain the geochemical behavior of lithophile, siderophile and chalcophile elements.	Geochemical comparison
8	Investigate the role of partition coefficients in the concentration of metals.	Numerical exercise and explanation
9	Compare genetic classifications used by Ridley, Robb and Pohl.	Comparative research paper
10	Investigate the relationship between plate tectonics and the global distribution of mineral deposits.	Global geological map
11	Select one major mineral deposit and reconstruct its geological history.	Genetic model
12	Investigate how multiple mineralization events can produce a composite orebody.	Geological evolution diagram
13	Evaluate how mineralogical, geochemical and isotopic evidence can be combined to identify ore genesis.	Evidence assessment
14	Prepare a scientific explanation of the source-transport-deposition-preservation model.	Illustrated presentation
15	Select a poorly understood or disputed mineral occurrence and formulate three competing genetic hypotheses.	Research proposal

Chapter 2: Magmatic Ore Deposits

No.	Review task	Expected output
1	Explain the relationship between partial melting and magmatic metal enrichment.	Scientific explanation
2	Compare fractional crystallization, magma mixing and assimilation.	Comparative diagram
3	Explain the formation of chromitite layers in layered intrusions.	Formation model
4	Investigate the geological setting and genesis of the Bushveld Complex.	Case-study report
5	Compare the Norilsk-Talnakh and Sudbury Ni-Cu-PGE systems.	Comparative genetic model
6	Explain the role of sulfur saturation in the formation of magmatic Ni-Cu deposits.	Process diagram
7	Calculate the influence of R-factor variations on metal concentration using a simplified sulfide-silicate partitioning dataset.	Numerical report
8	Investigate the geological characteristics of PGE-rich reefs.	Illustrated geological section
9	Explain how magmatic differentiation produces Fe-Ti-V oxide deposits.	Conceptual model
10	Investigate the origin of rare-metal pegmatites and explain the enrichment of Li, Cs and Ta.	Research report
11	Compare granite-related Sn-W deposits with pegmatite deposits.	Comparative table and diagram

No.	Review task	Expected output
12	Explain the formation of carbonatite-associated REE and Nb deposits.	Genetic model
13	Investigate the geological characteristics of kimberlite-hosted diamond deposits.	Geological case study
14	Construct a global distribution map of major magmatic ore deposit provinces.	Metallogenic map
15	Choose an unexplored or poorly understood magmatic system and formulate a geological model predicting the most likely mineralization styles.	Independent research project

Chapter 3: Hydrothermal Ore Deposits

No.	Review task	Expected output
1	Compare magmatic, meteoric, seawater, basinal and metamorphic fluids.	Comparative table
2	Explain the influence of temperature, pressure, salinity and pH on metal solubility.	Geochemical explanation
3	Interpret a hypothetical fluid inclusion dataset and reconstruct the evolution of a hydrothermal fluid.	Data interpretation report
4	Explain the formation of porphyry copper deposits from magma generation to metal precipitation.	Complete genetic cross-section
5	Compare low-, intermediate- and high-sulfidation epithermal systems.	Comparative genetic models
6	Reconstruct the formation of a skarn deposit using mineral assemblages and alteration relationships.	Geological model

No.	Review task	Expected output
7	Investigate the relationship between modern black smokers and ancient VMS deposits.	Research report
8	Compare Kuroko, Cyprus and Besshi-type VMS deposits.	Comparative geological report
9	Explain the origin of orogenic gold deposits and evaluate the evidence for metamorphic fluid sources.	Critical scientific review
10	Compare Carlin-type and orogenic gold deposits.	Comparative genetic model
11	Investigate the geological characteristics of MVT and SEDEX deposits.	Basin-scale cross-sections
12	Explain the formation of sediment-hosted copper deposits using redox and fluid-flow principles.	Formation diagram
13	Compare sandstone-hosted and unconformity-related uranium deposits.	Comparative case study
14	Evaluate two competing genetic models for an IOCG deposit.	Scientific argument supported by evidence
15	Select a real hydrothermal deposit and reconstruct its complete geological evolution using maps, mineralogy, geochemistry and published geochronology.	Research paper and oral defense

Chapter 4: Weathering and Supergene Ore Deposits

No.	Review task	Expected output
1	Explain the influence of climate on chemical weathering and mineral concentration.	Conceptual model
2	Construct a typical lateritic weathering profile and explain its vertical mineralogical zoning.	Geological cross-section
3	Compare residual enrichment and supergene enrichment.	Comparative report
4	Investigate the formation of bauxite deposits in tropical environments.	Case study
5	Explain nickel concentration during the weathering of ultramafic rocks.	Geochemical model
6	Compare different types of lateritic nickel deposits.	Comparative diagrams
7	Explain the formation of ion-adsorption REE deposits.	Process model
8	Reconstruct the evolution of an oxidized sulfide deposit.	Vertical zonation diagram
9	Explain the chemical processes responsible for gossan formation.	Geochemical report
10	Investigate the formation of secondary copper sulfide enrichment zones.	Geological cross-section
11	Compare primary and secondary copper mineral assemblages.	Mineralogical comparison
12	Explain the formation of roll-front uranium deposits.	Groundwater-flow and redox model

No.	Review task	Expected output
13	Investigate the relationship between groundwater movement and uranium precipitation.	Conceptual model
14	Evaluate how erosion and burial influence the preservation of ancient weathering profiles.	Geological interpretation
15	Select a real laterite or supergene deposit and reconstruct its weathering history, element mobility and enrichment mechanisms.	Independent research report

Chapter 5: Sedimentary Ore Deposits

No.	Review task	Expected output
1	Explain the physical principles of hydraulic sorting in placer formation.	Process diagram
2	Calculate and compare the settling behavior of different mineral grains using a simplified hydraulic dataset.	Numerical exercise
3	Investigate the formation of gold placer deposits in fluvial environments.	Geological report
4	Compare river, beach and paleoplacer deposits.	Comparative diagrams
5	Explain the formation and debated genesis of Witwatersrand gold deposits.	Critical research report
6	Reconstruct the formation of banded iron formations.	Geological and geochemical model
7	Explain the relationship between early ocean chemistry and BIF formation.	Earth-history diagram

No.	Review task	Expected output
8	Compare BIF and oolitic iron deposits.	Comparative report
9	Explain the formation of marine manganese deposits and nodules.	Process model
10	Investigate the chemical conditions required for evaporite formation.	Chemical precipitation sequence
11	Explain the formation of phosphorite deposits in marine environments.	Geological model
12	Investigate the role of microorganisms in sedimentary mineral concentration.	Research report
13	Compare chemical, biochemical and mechanical sedimentary concentration.	Comparative table
14	Construct a global distribution map of important sedimentary ore deposit provinces.	Metallogenic map
15	Select a sedimentary basin and develop a geological model predicting the most favorable environments for mineral concentration.	Basin evolution report and cross-section

Chapter 6: Metamorphic and Metamorphosed Ore Deposits

No.	Review task	Expected output
1	Explain the differences between metamorphic, metamorphosed and metamorphogenic deposits.	Classification report
2	Explain how pressure-temperature conditions influence mineral stability and metamorphic reactions.	P-T diagram

No.	Review task	Expected output
3	Investigate the generation of metamorphic fluids during dehydration and decarbonation.	Reaction and fluid-generation model
4	Explain how metamorphic fluids can transport and precipitate metals.	Fluid-flow diagram
5	Investigate the effects of regional metamorphism on VMS deposits.	Comparative mineralogical report
6	Reconstruct the original and metamorphic evolution of the Broken Hill Pb-Zn-Ag deposit.	Geological evolution model
7	Investigate the effects of metamorphism on the Kambalda Ni sulfide deposits.	Research report
8	Explain tungsten remobilization in the Felbertal scheelite deposit.	Geological process model
9	Investigate the relationship between regional metamorphism and orogenic gold mineralization in the Yilgarn Craton.	Scientific report
10	Compare metamorphic gold mineralization with magmatic-hydrothermal gold mineralization.	Comparative genetic model
11	Explain the metamorphic formation of graphite deposits.	Reaction model
12	Investigate the pressure-temperature conditions of kyanite, sillimanite and andalusite formation.	Mineral stability diagram
13	Compare metamorphic and metasomatic formation of industrial mineral deposits.	Comparative report
14	Evaluate geological evidence distinguishing pre-metamorphic mineralization from synmetamorphic mineralization.	Evidence-based interpretation

No.	Review task	Expected output
15	Select a real metamorphic or metamorphosed mineral deposit and reconstruct its original formation, metamorphic evolution and final orebody characteristics.	Independent research paper and oral presentation

6.1. How should the review tasks be assessed?

I recommend the following common assessment rubric.

Assessment criterion	Weight
Understanding of geological principles	20%
Interpretation of geological evidence	25%
Correctness of genetic model	25%
Scientific literature and independent research	15%
Presentation, figures and scientific communication	15%
Total	**100%**

For Tasks 15 in each chapter, require students to use at least five relevant scientific publications, including recent research papers where available.

The purpose is to teach students to develop defensible geological interpretations, not simply reproduce descriptions from textbooks.

Part VI. Detailed illustration program: 156 proposed figures

The following is the proposed figure register. Each figure has a title and a description of what it should communicate.

I recommend creating original diagrams for the conceptual and process figures, using properly licensed photographs and published datasets for real geological examples.

Chapter 1: 24 figures

FUNDAMENTALS OF ORE DEPOSITS

Figure	Proposed title and description
1.1	**Earth's Internal Structure and Distribution of Mineral Resources** – Cross-section showing the crust, mantle, core and major geological sources of elements.
1.2	**Mineral Occurrence, Resource and Ore Deposit** – Diagram distinguishing geological occurrence, resource and economically viable ore.
1.3	**Ore and Gangue Minerals in a Copper Ore Specimen** – Annotated photograph identifying valuable and non-valuable mineral phases.
1.4	**Relationship between Grade, Tonnage and Ore Value** – Conceptual graph explaining grade and tonnage.
1.5	**Major Ore Mineral Assemblages** – Photographic plate of common sulfide, oxide and native-element minerals.
1.6	**Mineral Associations in Different Deposit Types** – Comparative mineralogical assemblages.
1.7	**Primary, Secondary and Replacement Ore Textures** – Illustrated comparison of textural formation mechanisms.
1.8	**Typical Ore Structures** – Massive, disseminated, banded, breccia, vein and stockwork structures.
1.9	**Paragenetic Sequence of a Multistage Ore Deposit** – Mineral formation sequence through geological time.
1.10	**Distribution of Valuable Elements within Ore Minerals** – Diagram showing lattice substitution, inclusions and dispersed elements.
1.11	**Common Morphologies of Ore Bodies** – Three-dimensional shapes of tabular, lenticular, vein and pipe-like bodies.

Figure	Proposed title and description
1.12	**Orebody Geometry in Geological Cross-Sections** – Comparison of concordant and discordant mineralization.
1.13	**Structural Controls on Orebody Formation** – Faults, folds, shear zones and fracture systems.
1.14	**Strike, Dip and Plunge of Ore Bodies** – Stereographic and three-dimensional explanation.
1.15	**Comparison of Descriptive and Genetic Ore Deposit Classification** – Classification matrix.
1.16	**The Source-Transport-Deposition-Preservation Model** – Integrated ore-forming system.
1.17	**Major Geological Processes Responsible for Ore Formation** – Process flowchart linking magmatic, hydrothermal, sedimentary, weathering and metamorphic systems.
1.18	**Multistage Ore Formation and Overprinting** – Diagram showing successive mineralization and modification events.
1.19	**Geochemical Classification of Elements** – Periodic table highlighting lithophile, siderophile, chalcophile and atmophile elements.
1.20	**Element Partitioning between Melt, Fluid and Crystals** – Schematic of partition coefficients.
1.21	**Eh-pH Diagram and Mineral Stability** – Representative mineral stability fields.
1.22	**Global Tectonic Settings of Major Ore Deposits** – World tectonic map with major deposit environments.
1.23	**Methods for Investigating Ore Genesis** – Workflow from field observations to laboratory analyses and genetic modeling.

Figure	Proposed title and description
1.24	**Integration of Geological, Geochemical and Isotopic Evidence** – Example of multiple datasets contributing to one genetic model.

Chapter 2: 26 figures

MAGMATIC ORE DEPOSITS

Figure	Proposed title and description
2.1	**Major Sources of Magma in the Earth's Crust and Mantle** – Mantle and crustal melting processes.
2.2	**Partial Melting and Generation of Magma** – Diagram illustrating different degrees of partial melting.
2.3	**Magma Differentiation and Evolution** – Crystallization and chemical evolution.
2.4	**Fractional Crystallization of Mafic Magma** – Formation of mineral phases during cooling.
2.5	**Geochemical Evolution of Magmatic Systems** – Behavior of compatible and incompatible elements.
2.6	**Silicate-Sulfide Liquid Immiscibility** – Formation and separation of sulfide liquid droplets.
2.7	**Silicate-Oxide Liquid Immiscibility** – Formation of Fe-Ti oxide-rich melts.
2.8	**Sulfide Saturation in Mafic Magmas** – Effect of sulfur addition, magma contamination and cooling.
2.9	**The R-Factor in Magmatic Ni-Cu-PGE Deposits** – Illustration of magma-sulfide interaction.

Figure	Proposed title and description
2.10	**Formation of Chromite Layers in a Layered Mafic Intrusion** – Accumulation and crystallization processes.
2.11	**Stratiform Chromitite in the Bushveld Complex** – Geological photograph and interpreted stratigraphic section.
2.12	**Podiform Chromite in Ophiolitic Rocks** – Field photograph and tectonic context.
2.13	**Formation of Komatiite-Hosted Nickel Sulfide Deposits** – Lava flow and sulfide accumulation model.
2.14	**Conduit-Hosted Magmatic Ni-Cu Sulfide Deposits** – Magma conduit and sulfide accumulation.
2.15	**Norilsk-Talnakh Ni-Cu-PGE Deposit** – Geological cross-section and intrusive architecture.
2.16	**Sudbury Impact-Related Ni-Cu-PGE System** – Schematic of the Sudbury structure and mineralized rocks.
2.17	**PGE-Rich Reefs in Layered Intrusions** – Stratigraphic position and mineralization distribution.
2.18	**Fe-Ti-V Oxide Mineralization in Mafic-Anorthositic Systems** – Oxide accumulation model.
2.19	**Evolution of Rare-Metal Pegmatite Systems** – Zoned pegmatite formation.
2.20	**Internal Zonation of a Li-Cs-Ta Pegmatite** – Mineral distribution from margin to core.
2.21	**Granite-Related Sn-W Mineralization** – Granite intrusion, greisen and associated veins.
2.22	**Carbonatite Magmatism and REE-Nb Mineralization** – Carbonatite emplacement and mineral concentration.

Figure	Proposed title and description
2.23	**Bayan Obo Rare-Earth Deposit** – Regional setting and geological architecture, with competing genetic interpretations.
2.24	**Kimberlite Magma Ascent and Diamond Preservation** – Mantle source, rapid ascent and emplacement.
2.25	**Global Distribution of Major Magmatic Ore Deposits** – Locations of major chromite, Ni-Cu-PGE, pegmatite and carbonatite deposits.
2.26	**Comparative Magmatic Ore-Forming Models** – Summary of the major magmatic concentration mechanisms.

Chapter 3: 40 figures

HYDROTHERMAL ORE DEPOSITS

Figure	Proposed title and description
3.1	**Components of a Hydrothermal System** – Heat source, fluid source, fluid pathways and mineralization zones.
3.2	**Sources of Hydrothermal Fluids** – Magmatic, meteoric, marine, basinal and metamorphic fluid sources.
3.3	**Hydrothermal Fluid Composition** – Major chemical components and dissolved metals.
3.4	**Metal Complex Formation in Hydrothermal Fluids** – Illustrations of chloride and bisulfide complexes.
3.5	**Fluid Flow through Faults and Fractures** – Structural controls on permeability and hydrothermal migration.
3.6	**Mechanisms of Metal Precipitation** – Cooling, boiling, pressure reduction, mixing and fluid-rock reaction.

Figure	Proposed title and description
3.7	**Hydrothermal Alteration Zoning** – Potassic, phyllic, propylitic and argillic alteration.
3.8	**Generalized Porphyry Copper Deposit Cross-Section** – Intrusion, stockwork veins, sulfides and alteration zones.
3.9	**Evolution of a Porphyry Magmatic-Hydrothermal System** – Magma emplacement and fluid exsolution.
3.10	**Porphyry Copper Vein Types** – Comparative photographs of major vein generations.
3.11	**Alteration Zoning in a Porphyry Copper Deposit** – Mineral assemblages and their spatial relationships.
3.12	**Escondida Porphyry Copper Deposit** – Geological setting and representative mineralization model.
3.13	**Low-, Intermediate- and High-Sulfidation Epithermal Systems** – Comparative geological cross-sections.
3.14	**Boiling and Gold Precipitation in Epithermal Systems** – Fluid pressure and temperature effects.
3.15	**Crustal Level of Epithermal Mineralization** – Relationship with shallow magmatic-hydrothermal systems.
3.16	**Skarn Formation around an Intrusive Body** – Contact zone and metasomatic replacement.
3.17	**Prograde and Retrograde Skarn Evolution** – Changes in mineral assemblages.
3.18	**Mineralogical Zoning in Skarn Deposits** – Garnet, pyroxene, amphibole and sulfide distributions.
3.19	**Greisen Formation in Granite Systems** – Fluid migration and quartz-muscovite alteration.

Figure	Proposed title and description
3.20	**Albitization and Albitite-Related Mineralization** – Sodium metasomatism and alteration pathways.
3.21	**Carbonate-Hosted Replacement Mineralization** – Fluid migration and sulfide replacement.
3.22	**Polymetallic Vein Systems** – Structural controls and metal zoning.
3.23	**IOCG Deposit Geological Model** – Iron oxide-rich hydrothermal mineralization.
3.24	**Iron Oxide-Apatite Mineralization** – Magnetite-apatite association.
3.25	**Magmatic-Hydrothermal Systems in Alkaline Complexes** – REE and Nb concentration.
3.26	**Modern Seafloor Black Smoker System** – Seawater circulation and hydrothermal discharge.
3.27	**Formation of a VMS Deposit** – Submarine volcanic rocks and sulfide accumulation.
3.28	**Kuroko-Type VMS Mineral Zoning** – Typical sulfide mineral assemblages.
3.29	**Cyprus-Type VMS Deposit** – Ophiolite setting and mineralization.
3.30	**Ancient VMS Deposit Compared with a Modern Hydrothermal Vent** – Geological analogy.
3.31	**Orogenic Gold Deposit Geological Model** – Deformation, shear zones and quartz-carbonate veins.
3.32	**Generation and Migration of Metamorphic Fluids** – Dehydration reactions and fluid pathways.
3.33	**Gold Transport in Aqueous-Carbonic Fluids** – Schematic metal complexation and deposition.

Figure	Proposed title and description
3.34	**Carlin-Type Gold Deposit Model** – Sedimentary host rocks, alteration and disseminated gold.
3.35	**Orogenic and Carlin-Type Gold Comparison** – Structural and mineralogical differences.
3.36	**Sedimentary Basin Brine Circulation** – Basin architecture and fluid migration.
3.37	**MVT Lead-Zinc Deposit Formation** – Carbonate replacement by basinal fluids.
3.38	**SEDEX Deposit Formation** – Basin-floor hydrothermal discharge and sulfide accumulation.
3.39	**Red-Bed Copper Mineralization** – Oxidized and reduced sedimentary environments.
3.40	**Sandstone-Hosted and Unconformity-Related Uranium Deposits** – Comparative fluid-flow and redox models.

Chapter 4: 20 figures

WEATHERING AND SUPERGENE ORE DEPOSITS

Figure	Proposed title and description
4.1	**Chemical Weathering of Rocks** – Dissolution, oxidation and hydrolysis.
4.2	**Typical Lateritic Weathering Profile** – Soil, laterite, saprolite and bedrock.
4.3	**Climate and Weathering Intensity** – Comparison of tropical, arid and temperate weathering.
4.4	**Element Mobility during Weathering** – Relative mobility of important elements.
4.5	**Residual Enrichment Process** – Removal of mobile components and residual concentration.

Figure	Proposed title and description
4.6	**Bauxite Deposit Formation** – Development of aluminum-rich lateritic profiles.
4.7	**Lateritic Nickel Deposit Formation** – Vertical nickel concentration in weathered ultramafic rocks.
4.8	**Mineralogical Zoning in Nickel Laterites** – Limonite, transition zone and saprolite.
4.9	**Ion-Adsorption REE Deposit Formation** – REE release and adsorption on clay minerals.
4.10	**Weathering-Related Iron and Manganese Enrichment** – Oxide precipitation in regolith.
4.11	**Oxidation of Sulfide Ore Bodies** – Development of the oxidation zone.
4.12	**Formation of Gossan** – Oxidation and residual accumulation of iron oxides.
4.13	**Vertical Zoning of a Supergene Copper Deposit** – Oxide, enrichment and primary sulfide zones.
4.14	**Secondary Copper Sulfide Enrichment** – Downward metal migration and precipitation.
4.15	**Chalcocite and Covellite Formation** – Mineralogical evidence of secondary enrichment.
4.16	**Groundwater Movement and Redox Boundaries** – Hydrological controls on mineral precipitation.
4.17	**Roll-Front Uranium Deposit Formation** – Oxidized and reduced groundwater zones.
4.18	**Calcrete-Hosted Uranium Mineralization** – Evaporation and carbonate-rich groundwater.
4.19	**Preservation of Ancient Weathering Profiles** – Burial, erosion and tectonic evolution.

Figure	Proposed title and description
4.20	**Global Distribution of Major Laterite, Bauxite and Supergene Deposits** – World distribution map.

Chapter 5: 22 figures

SEDIMENTARY ORE DEPOSITS

Figure	Proposed title and description
5.1	**Principles of Sediment Transport and Deposition** – Sediment movement and accumulation.
5.2	**Hydraulic Sorting of Heavy Minerals** – Effects of grain density, size and water velocity.
5.3	**Relationship between Grain Size, Density and Settling Velocity** – Conceptual and quantitative diagram.
5.4	**Sedimentary Basin Evolution and Mineralization** – Basin development and depositional environments.
5.5	**Fluvial Placer Gold Formation** – Gold accumulation in river channels and gravel bars.
5.6	**Beach Placer Deposit Formation** – Wave action and heavy-mineral sorting.
5.7	**Ancient Paleoplacer Formation** – Conglomerate-hosted gold and uranium concentration.
5.8	**Witwatersrand Basin and Gold Mineralization** – Regional geological context and debated genetic models.
5.9	**Heavy-Mineral Placer Assemblages** – Gold, cassiterite, zircon, rutile and ilmenite.
5.10	**Formation of Banded Iron Formations** – Marine iron precipitation and sedimentation.

Figure	Proposed title and description
5.11	**BIF Mineralogical and Textural Characteristics** – Hematite, magnetite, chert and banding.
5.12	**Evolution of Ocean Chemistry and Iron Deposition** – Changes through geological time.
5.13	**Oolitic Ironstone Formation** – Formation and accumulation of iron-rich ooids.
5.14	**Sedimentary Manganese Deposit Formation** – Chemical precipitation and sedimentary concentration.
5.15	**Marine Manganese Nodules and Crusts** – Ocean-floor accumulation.
5.16	**Evaporite Formation and Mineral Precipitation Sequence** – Evaporation and progressive brine concentration.
5.17	**Phosphorite Formation in Marine Environments** – Upwelling and phosphorus cycling.
5.18	**Microbial Processes in Sedimentary Mineralization** – Biological mediation of chemical reactions.
5.19	**Organic-Rich Sediments and Metal Accumulation** – Relationship between organic matter and U, V, Ni and Mo.
5.20	**Chemical, Biochemical and Mechanical Sedimentary Ore Formation** – Comparative models.
5.21	**Evolution of Sedimentary Metallogeny through Geological Time** – Timeline of important sedimentary deposit types.
5.22	**Global Distribution of Major Sedimentary Ore Deposits** – World map showing important BIF, placer, Mn and phosphate regions.

Chapter 6: 24 figures

METAMORPHIC AND METAMORPHOSED ORE DEPOSITS

Figure	Proposed title and description
6.1	**Classification of Metamorphic, Metamorphosed and Metamorphogenic Deposits** – Conceptual classification diagram.
6.2	**Major Metamorphic Environments** – Regional, contact, dynamic and subduction-related metamorphism.
6.3	**Metamorphic Facies and Pressure-Temperature Conditions** – Facies diagram with representative mineral assemblages.
6.4	**Dehydration and Decarbonation Reactions during Metamorphism** – Generation of metamorphic fluids.
6.5	**Metamorphic Fluid Migration through Shear Zones** – Fluid flow and structural focusing.
6.6	**Metal Transport in Metamorphic Fluids** – Conceptual model of metal mobilization and precipitation.
6.7	**Pressure-Temperature-Time Evolution of a Metamorphic Ore System** – P-T-t path and mineralization stages.
6.8	**Mineralogical and Textural Changes in Metamorphosed Sulfide Ores** – Recrystallization and remobilization.
6.9	**Metamorphosed VMS Deposit Geological Model** – Original mineralization and subsequent metamorphic modification.
6.10	**Metamorphic Remobilization of Sulfide Minerals** – Metal redistribution and vein formation.
6.11	**Metamorphosed Ni-Cu-PGE Deposits** – Comparison of primary and modified sulfide textures.
6.12	**Metamorphic Evolution of Banded Iron Formation** – Recrystallization and metamorphic iron enrichment.

Figure	Proposed title and description
6.13	**Metamorphic Remobilization and Formation of New Ore Concentrations** – Comparison of pre-existing and newly precipitated mineralization.
6.14	**Orogenic Gold and Regional Metamorphism** – Relationship between deformation, fluids and gold mineralization.
6.15	**Tungsten Remobilization in the Felbertal Scheelite Deposit** – Geological setting and mineralization model.
6.16	**Gold Mineralization in High-Grade Metamorphic Terranes** – Amphibolite- and granulite-facies examples.
6.17	**Metamorphic Graphite Formation** – Transformation of carbonaceous material.
6.18	**Kyanite, Sillimanite and Andalusite Stability Fields** – P-T phase relationships.
6.19	**Talc and Magnesite Formation in Metamorphosed Ultramafic Rocks** – Fluid-rock reaction.
6.20	**Contact Metamorphic Industrial Mineral Formation** – Wollastonite and garnet-bearing assemblages.
6.21	**Broken Hill Pb-Zn-Ag Deposit** – Geological cross-section and metamorphic architecture.
6.22	**Kambalda Ni Sulfide Deposits** – Komatiite-hosted mineralization and metamorphic overprint.
6.23	**Yilgarn Craton Gold Mineralization** – Structural and metamorphic geological context.
6.24	**Integrated Model of Metamorphic Ore Deposit Evolution** – Original mineralization, metamorphism, fluid flow, remobilization and final preservation.

(156 core figures proposed)

6.2. Important guidelines for producing the figures

I recommend the following production standards:

1. All conceptual ore-forming models should have clear arrows indicating fluid, magma, metal and sediment movement.
2. Geological cross-sections should include depth, geological structures, host rocks, mineralized zones and alteration zones.
3. World maps should include tectonic boundaries, metallogenic provinces and representative deposit locations.
4. Microscopic images should include mineral names, scale bars and descriptions of the diagnostic textures.
5. Comparative figures should use consistent geological symbols and legends throughout the textbook.
6. Every figure should have a caption explaining its scientific significance, not merely identifying the objects in the photograph.

For real deposit photographs and geological maps, use original field photographs, public-domain geological survey material, open-access scientific publications or images for which you have obtained permission.

Do not copy the figures from Ridley, Robb or other copyrighted books without appropriate permission.

Part VII. Additional references and downloadable research materials

Besides the six books you have already used, I recommend expanding your reference library with the following publications and resources.

I have separated them into three categories: fundamental textbooks, specialized scientific publications and free teaching/research materials.

7.1. Important additional textbooks

No.	Reference	Main use in your textbook
1	Guilbert & Park (1986), *The Geology of Ore Deposits*	General principles, orebody geology, deposit models and geological controls

No.	Reference	Main use in your textbook
2	F. Pirajno (2009), *Hydrothermal Processes and Mineral Systems*	Hydrothermal processes, fluid migration and mineral systems
3	K. H. Misra (2000), *Understanding Mineral Deposits*	Deposit classification and geological characteristics
4	A. M. Evans (1993), *Ore Geology and Industrial Minerals: An Introduction*	General deposit characteristics and industrial mineral deposits
5	Laurence Robb (2020), *Introduction to Ore-Forming Processes*, 2nd ed.	Modern ore-forming mechanisms
6	Walter L. Pohl (2020), *Economic Geology: Principles and Practice*, 2nd ed.	Metamorphic deposits, metallogenesis and classification
7	F. Neukirchen & G. Ries (2020), *The World of Mineral Deposits*	Geological illustrations and global examples
8	F. M. Vokes, B. Marshall & P. G. Spry (eds.), *Metamorphosed and Metamorphogenic Ore Deposits*	Detailed metamorphic ore deposit research

For the last reference, I would make a special effort to obtain the complete SEG publication.

7.2. Free or officially downloadable PDF research resources

These are particularly useful because they provide geological models, scientific data and illustrations that can support your textbook.

1. USGS Bulletin 1693 – Mineral Deposit Models

Cox, D. P. and Singer, D. A. (1986).

Contains 85 descriptive deposit models and 60 grade-tonnage models.

Useful for writing Chapters 1–5 and developing comparative deposit models.

(Free official PDF)

Download PDF

USGS publication page

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2. USGS Porphyry Copper Deposit Model

Ayuso et al. (2010).

Excellent for porphyry copper geology, alteration, mineralization and genetic models.

(Free official PDF)

Access report and PDF

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3. USGS – Some Industrial Mineral Deposit Models

Orris and Bliss (1991).

Particularly useful for Chapter 6's metamorphic industrial minerals and Chapter 4.

(Free official resource)

Access report

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4. Reviews in Economic Geology, Volume 11 – Metamorphosed and Metamorphogenic Ore Deposits

The free product-information PDF includes the full table of contents. The complete 310-page PDF is available for purchase from SEG.

Download free contents PDF

Obtain complete PDF

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5. Cambridge University Press – Ore Deposit Geology

John Ridley (2013).

Cambridge provides an official downloadable frontmatter PDF, including introductory publication material, but not the complete book for free.

Download official frontmatter PDF

Book access

6. The Geology of Ore Deposits

Thomas and MacAlister (1909).

A historical public-domain textbook with a complete chapter on metamorphic ore deposits.

Access historical PDF

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7. Engaging Engineering Students in Geoscience Through Case Studies and Active Learning

Holley (2017).

A very useful paper for designing geological case-study assignments and active learning.

(Free research PDF)

Download research paper

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8. Geology and Environment: A Problem-Based Learning Study in Higher Education

2021, *Geosciences*.

Useful for designing the PBL methodology and evaluating learning outcomes.

Open-access research paper

7.3. University lecture notes and teaching resources

These are especially useful for studying how international lecturers organize their teaching materials.

University/resource	Recommended use	Link
University of Geneva – Ore Deposits Introductory Course	Lecture notes, deposit classification, hydrothermal and metamorphic deposits	Access course
University of Oslo – Ore-Forming Processes	Practical microscopy, mineral chemistry and field-based teaching	Access course
University of Georgia – Economic Geology	Learning outcomes, interpretation at multiple scales and field assignments	Access course

University/resource	Recommended use	Link
National Association of Geoscience Teachers	Case-based economic geology teaching	Pebble Alaska case study

7.4. Global mineral deposit maps and geological databases

For your textbook, global distribution maps will be very important.

I particularly recommend the following resource.

Fennoscandian Mineral Deposits Database and Maps

Geological Survey of Finland and partner geological surveys.

Provides metallogenic maps, deposit databases, downloadable geographic datasets and a detailed explanatory publication.

Access maps and data

I would use these resources to develop:

- Global distribution of porphyry copper deposits.
- Distribution of Ni-Cu-PGE deposits.
- Global chromite provinces.
- Distribution of VMS and SEDEX deposits.
- Major orogenic gold provinces.
- Distribution of MVT and sediment-hosted copper deposits.
- Major laterite and bauxite regions.
- Global BIF provinces.
- Distribution of metamorphosed massive sulfide deposits.
- Major metamorphic and metamorphogenic mineralization regions.

For the maps, make sure that the underlying geological datasets and deposit classifications are properly cited.

Part VIII. How to implement the textbook project

I recommend treating the preparation of this textbook as a research and curriculum-development project, rather than simply rewriting the existing manuscript.

8.1. Six stages of textbook development

1. **Stage 1. International literature review**

   Collect the important textbooks, university lecture notes, scientific reviews and recent research papers.

   Main output: A literature database organized by the six chapters.

2. **Stage 2. Finalize the scientific structure**

   Approve the detailed contents, genetic classification, chapter relationships and learning objectives.

   Main output: Final textbook table of contents.

3. **Stage 3. Develop geological figures**

   Produce original conceptual diagrams, global maps, geological cross-sections, mineralogical photographs and geochemical illustrations.

   Main output: A figure database containing at least 156 core figures.

4. **Stage 4. Write the scientific manuscript**

   Write each subsection using the common teaching structure: geological problem, theory, evidence, mechanism, case study and thinking problems.

   Main output: Six research-based chapters.

5. **Stage 5. Develop PBL activities**

   Prepare case studies, datasets, student assignments, model answers, group activities and project evaluation rubrics.

   Main output: Student learning activities and an instructor's guide.

6. **Stage 6. Scientific and educational review**

   Invite economic geologists, petrologists, geochemists and university teaching specialists to review the manuscript.

   Main output: Final textbook manuscript and supplementary teaching materials.

8.2. Six integrated semester-long PBL projects

In addition to the individual thinking problems and 90 review tasks, I recommend six larger projects, one associated with each chapter.

Chapter	Project title	Main student activity
1	What Makes a Mineral Occurrence an Ore Deposit?	Analyze geological, mineralogical and geochemical evidence and classify an unknown deposit
2	Why Are Ni-Cu-PGE Metals Concentrated in Certain Magmatic Systems?	Investigate magma differentiation, sulfide saturation and metal partitioning
3	Reconstructing the Evolution of a Hydrothermal Ore System	Integrate fluid inclusions, alteration, isotopes and mineralization stages
4	Reconstructing an Ancient Weathering Profile	Explain residual enrichment and supergene processes using a geological profile
5	Where and Why Do Valuable Minerals Accumulate in Sedimentary Basins?	Interpret sedimentary facies, hydraulic sorting and chemical precipitation
6	Was the Ore Deposit Formed Before or During Metamorphism?	Compare original mineralization and metamorphic modification using geological evidence

8.3. Example of a complete PBL assignment

(Example PBL assignment · Chapter 3)

Reconstructing the Genesis of an Unknown Hydrothermal Deposit

Problem scenario

A geological research team has discovered an area with the following characteristics:

- A felsic intrusive complex.

- Quartz-sulfide veins.
- Disseminated chalcopyrite and pyrite.
- Potassic and phyllic alteration.
- Elevated copper and molybdenum concentrations.
- Several generations of cross-cutting veins.

However, the geological history of the mineralization is not yet established.

Student investigation

1. Identify the possible deposit types.
2. Develop at least two competing genetic hypotheses.
3. Identify the most useful additional geological evidence.
4. Interpret hypothetical fluid inclusion and isotope datasets.
5. Reconstruct the hydrothermal evolution.
6. Draw a geological cross-section.
7. Explain the most likely metal precipitation mechanisms.
8. Defend the final interpretation.

Expected deliverables

- Geological research report.
- Annotated geological cross-section.
- Hydrothermal alteration map.
- Mineralization sequence.
- Genetic model.
- Ten-minute scientific presentation.

Assessment

Evaluate the geological reasoning, quality of evidence, genetic model, consideration of alternative hypotheses and scientific communication.

The advantage of this assignment is that students must use knowledge from several subsections of the same chapter rather than answer an isolated question.

Part IX. My final recommendations for making this textbook internationally competitive

I would pay particular attention to the following ten improvements.

1. **Use a consistent genetic classification.** Explain the relationship between geological processes, geological environments and deposit types.

2. **Give more attention to ore-forming mechanisms.** Explain why metals become concentrated, how they are transported and why they precipitate.

3. **Integrate geological evidence into theoretical explanations.** Every major deposit type should have examples of actual geological observations.

4. **Develop metamorphic deposits as a complete chapter.** Clearly distinguish metamorphic, metamorphosed and metamorphogenic deposits.

5. **Include quantitative geological explanations.** Introduce partition coefficients, mass balance, fluid chemistry, phase equilibria and geochemical diagrams at an appropriate undergraduate level.

6. **Use global case studies.** Do not restrict the examples to one country or geological region. Include Africa, Asia, Europe, North America, South America and Australia.

7. **Develop original geological illustrations.** A good genetic diagram often explains a geological mechanism better than several pages of text.

8. **Include research questions throughout the chapters.** Encourage students to interpret evidence and consider scientific uncertainty.

9. **Separate student materials from instructor materials.** Provide full model answers, additional datasets, suggested reading and assessment rubrics in an instructor's guide or supplementary online materials.

10. **Connect all six chapters through geological evolution.** Students should understand that mineral deposits are not isolated geological objects; they are products of Earth's evolving magmatic, hydrothermal, sedimentary, surface and metamorphic systems.

9.1. Recommended supplementary textbook materials

In addition to the 368-page textbook, I recommend preparing four supplementary resources.

Supplementary resource	Suggested content
Student workbook	Geological interpretation exercises, datasets, thinking problems and review tasks
Instructor's manual	Model answers, PBL instructions, grading rubrics and teaching schedules
Geological image atlas	High-resolution ore specimens, photomicrographs, cross-sections and global maps
Digital research database	Links to scientific publications, open datasets, geological maps and real deposit case studies

This would make the textbook much more useful for university teaching.

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10. Suggested writing sequence

Although the six chapters are now outlined, I would not recommend writing them sequentially without first establishing a consistent scientific framework.

My suggested order is:

Priority	Work	Reason
1	Chapter 1	Establish terminology, classification and theoretical principles
2	Chapter 2	Develop the first complete process-based deposit models
3	Chapter 3	Establish the most extensive fluid-related genetic framework
4	Chapter 6	Develop the specialized metamorphic classification and avoid duplication with Chapter 3

Priority	Work	Reason
5	Chapter 4	Develop weathering and supergene processes
6	Chapter 5	Complete sedimentary mineralization and its relationship to the other systems

After Chapters 1–3 and 6 have been scientifically coordinated, Chapters 4 and 5 can be developed with a clearer classification framework.

10.2. The first manuscript I recommend developing

I suggest beginning with Chapter 1 in full manuscript form.

It should contain:

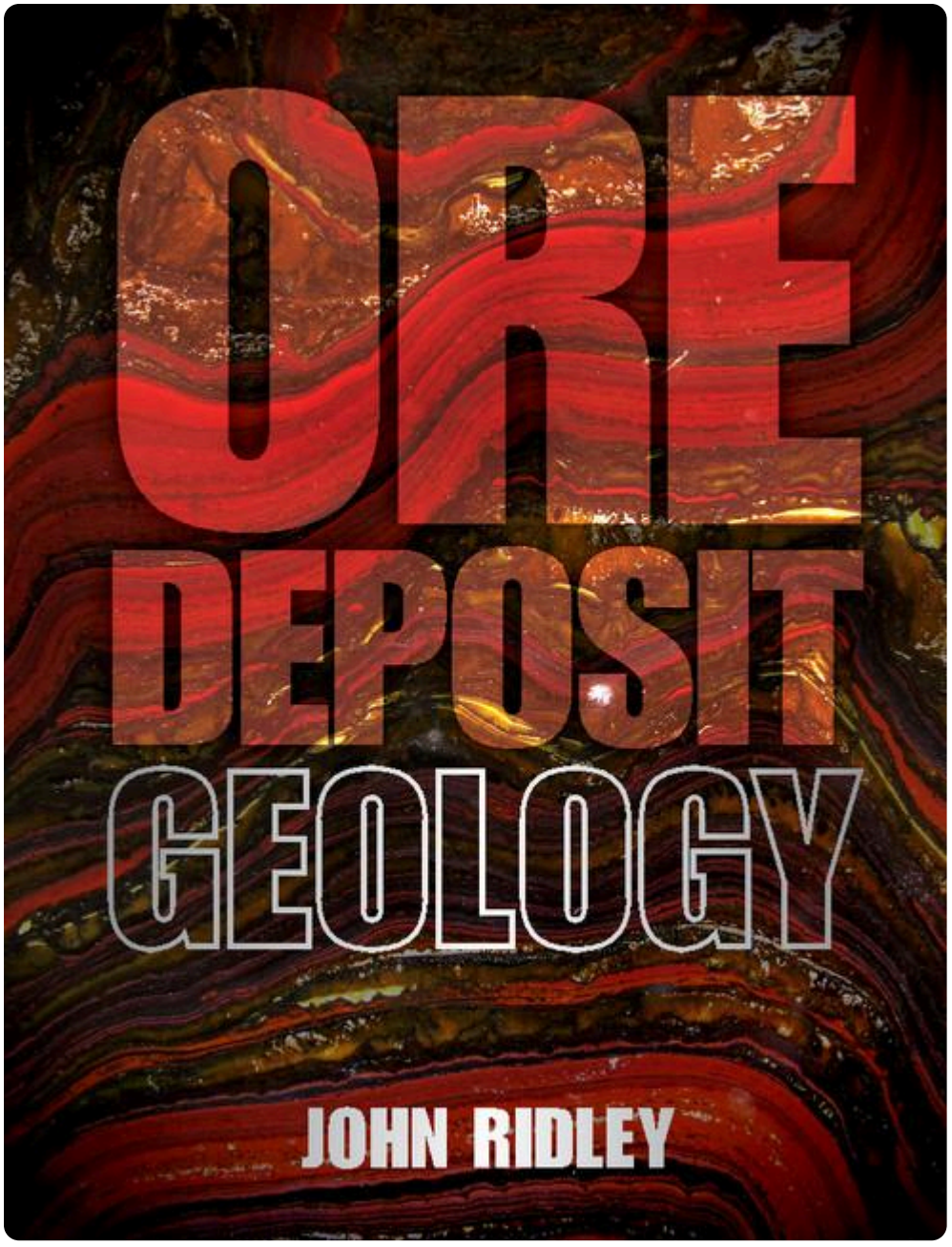
- Complete scientific explanations of all subsections.
- Approximately 46 pages of textbook material.
- Approximately 24 original figures.
- Geological examples and case studies.
- Thinking problems and scientifically reasoned answers.
- A chapter-level PBL assignment.
- Fifteen review tasks.
- References for every major scientific topic.

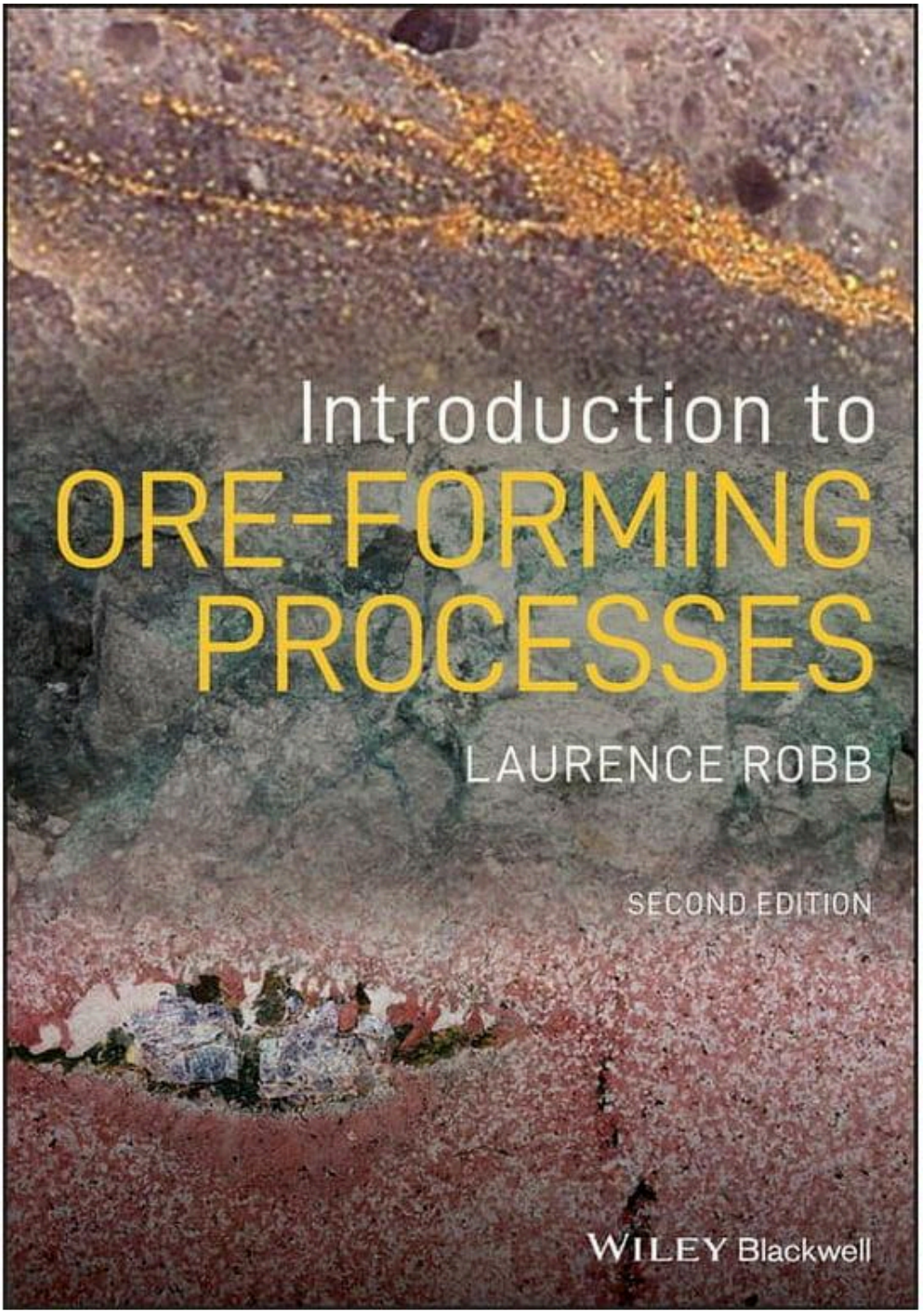
Once the style, scientific depth, figure format and PBL structure have been approved, the same writing standards can be applied to the other five chapters.

My overall assessment: Your original textbook already has a sound foundation in the traditional classification of mineral deposits. Its greatest opportunity for improvement is to move from a mainly descriptive classification of deposit types to a modern explanation of geological processes, supported by actual research evidence and student investigation.

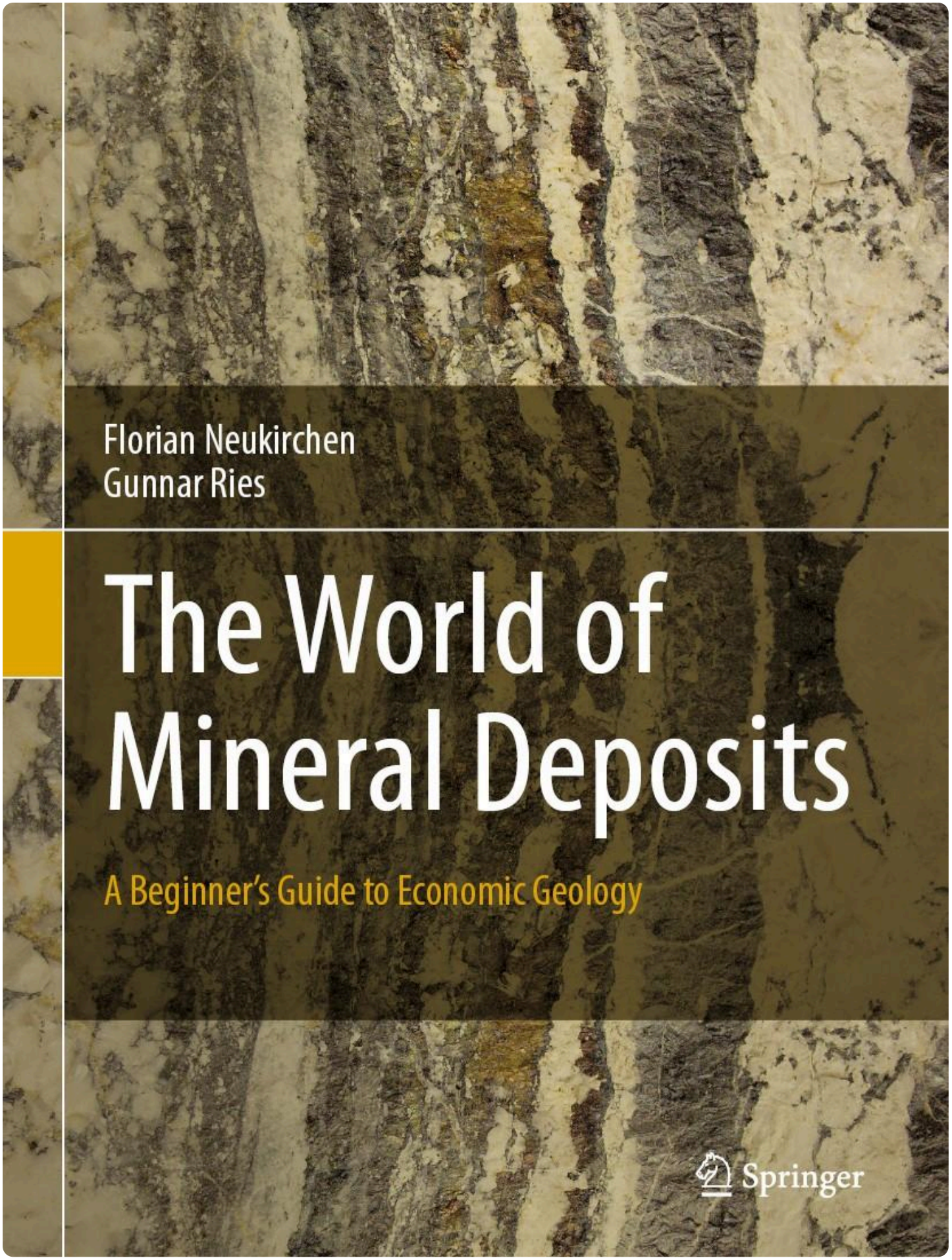
The proposed six-chapter structure can achieve that objective while retaining the identity of your existing subject and avoiding unnecessary duplication of mineral exploration, mining and mineral processing courses.

I recommend developing the textbook as a complete teaching package: **a 368-page core textbook, at least 156 original or properly licensed figures, 90 chapter review tasks, six integrated PBL projects and a separate instructor's guide.**

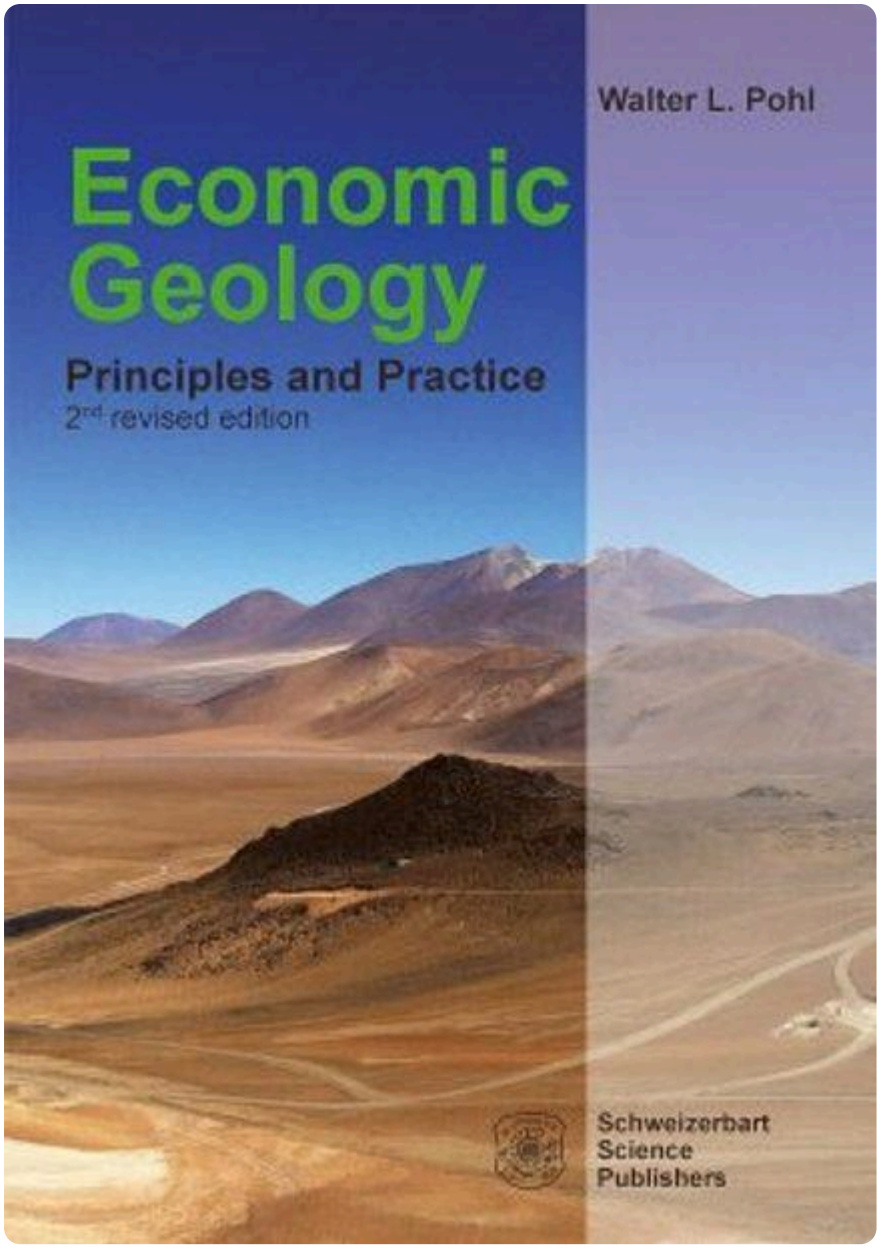




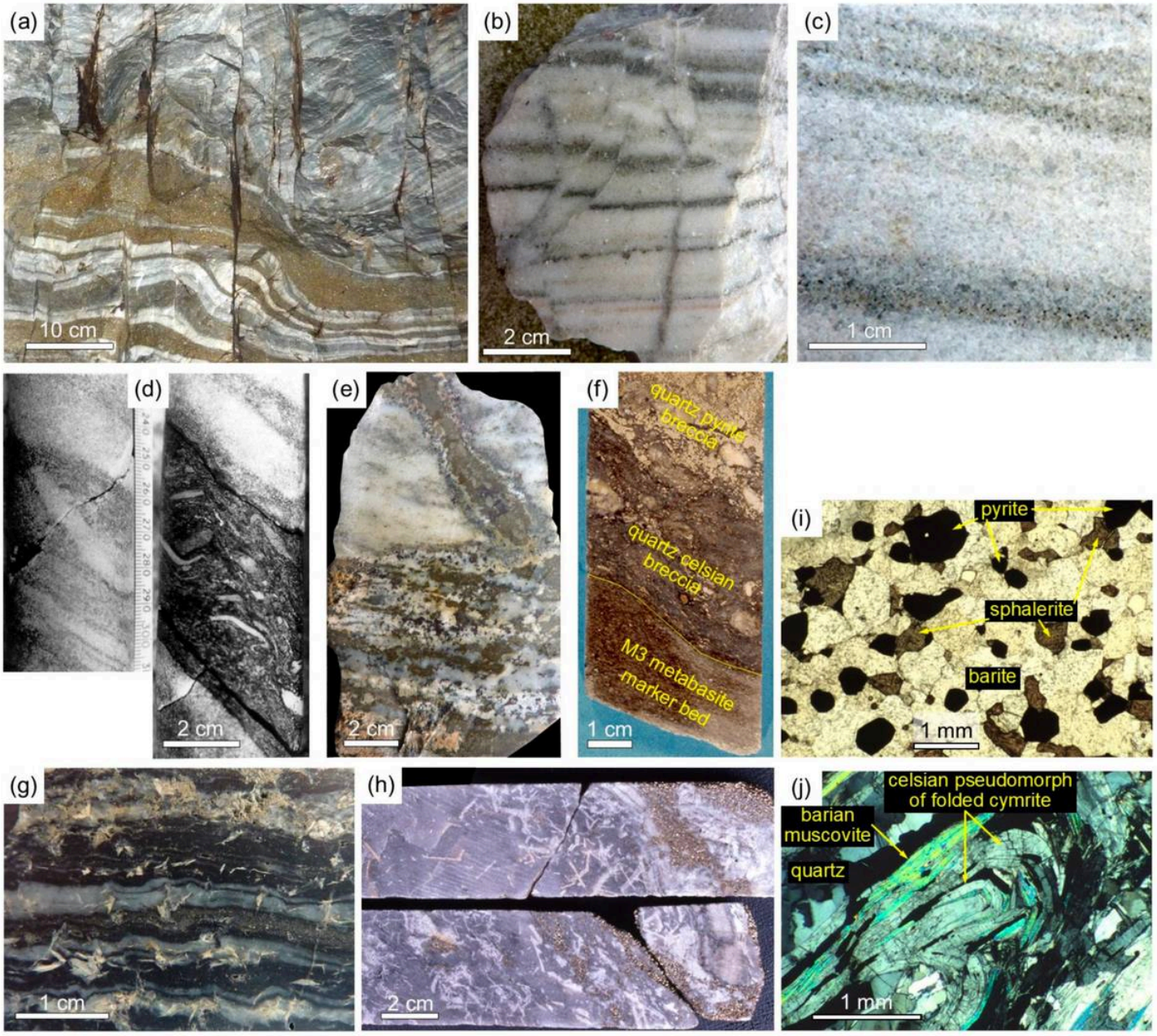
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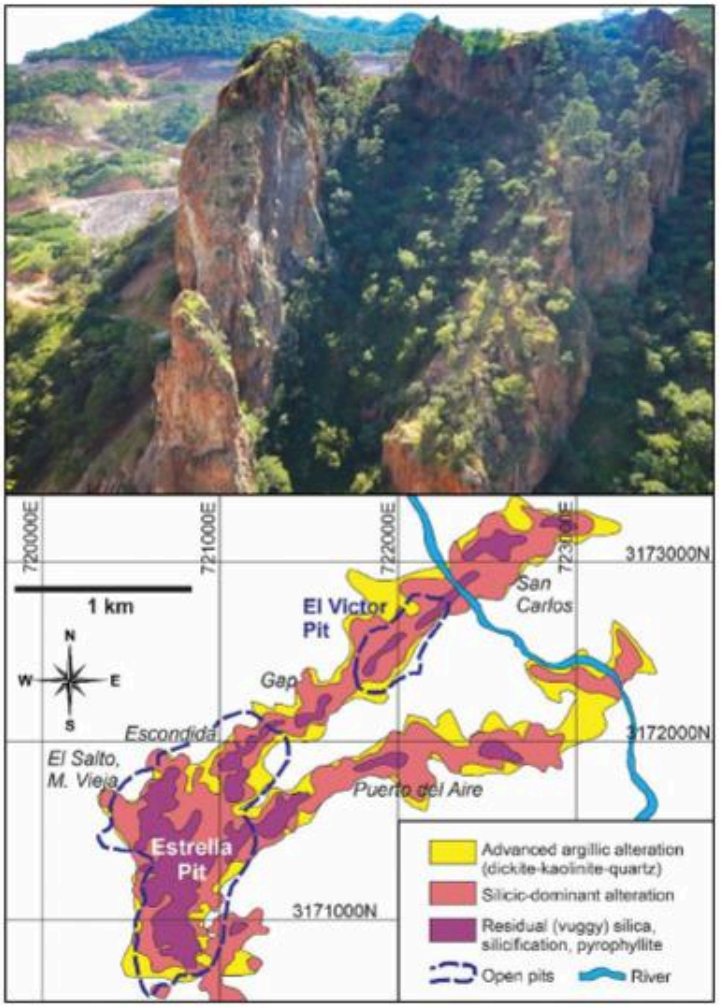

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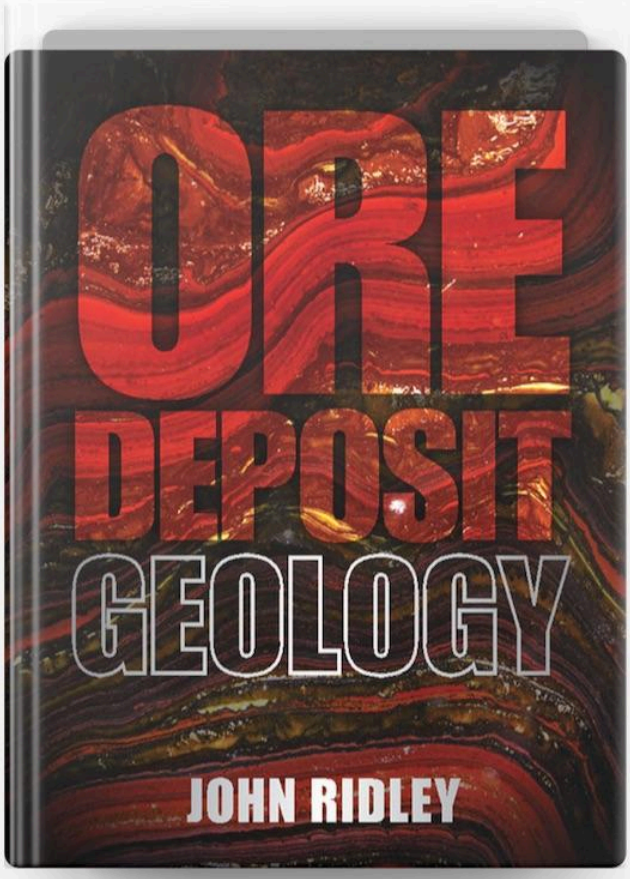
APPLIED STRUCTURAL GEOLOGY OF ORE-FORMING HYDROTHERMAL SYSTEMS



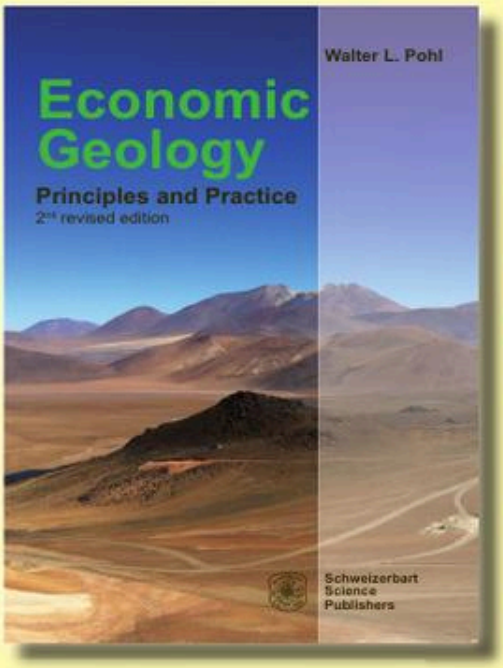
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Walter L. Pohl

Economic Geology

Principles and Practice

2020. 2nd revised edition.
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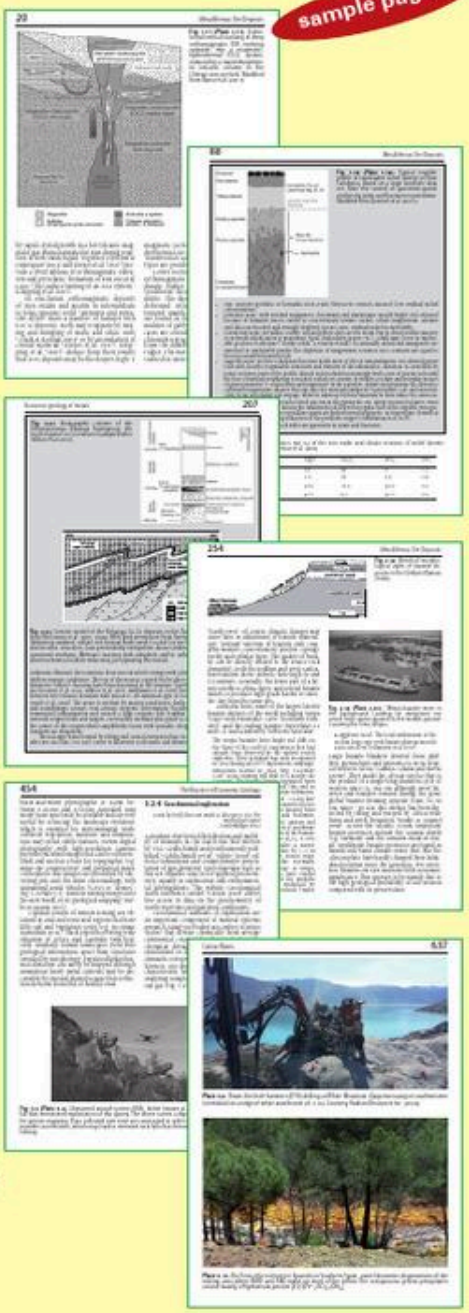
Since the 1st edition of this book was published, both science and practice of economic geology have advanced in great strides. Improvements in understanding the Earth's process systems that form raw material deposits are reflected in this revised second edition. The scientific ambition of the book is to place the extraordinary variability of mineral deposits into the framework of fundamental petrogenetic-geological process systems.

The book covers the entire field of geology applied to mineral deposits, including industrial minerals, coal and hydrocarbons. Illuminating insights can be gained from this; a vivid example for this are organic matter-rich sediments that are foremost source rocks for conventional hydrocarbons, but also host unconventional oil and gas, and metal accumulations. Numerous individual mineral deposits are presented as practical examples, to discuss reserve figures, ore grade, origin and geological characteristics. The principles of searching for ores and minerals (exploration), the study and valuation of newly found deposits, and environmental issues are treated in a separate chapter. The overview character of this textbook requires waiving much detail but for compensation, ample cross-references and references to web and printed sources are provided.

The number of humans on our globe continues to increase and standards of living improve rapidly. Both contribute to a steadily rising consumption of raw materials. Economic geology is a central actor in satisfying this growing demand by exploration and mining, and by mediating social and environmental impacts. The text also covers related issues that arise during the life cycle of a mine and after its closure, with an emphasis on sustainable and 'green' mining.

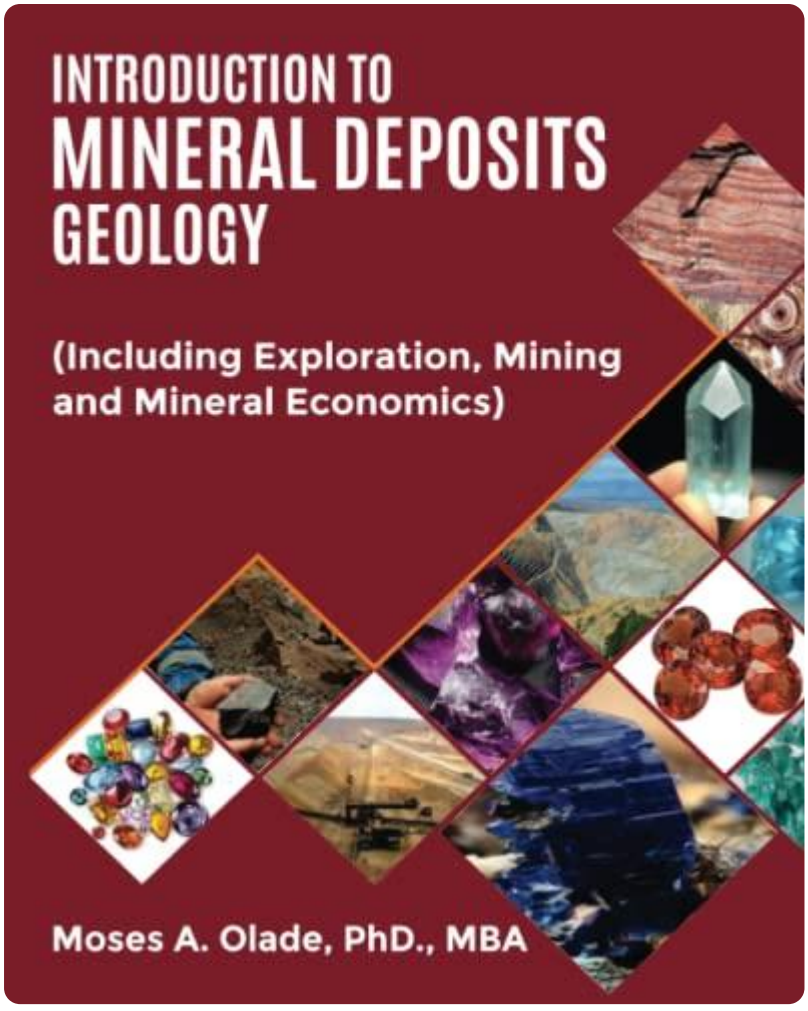
Students and teachers of economic geology and related disciplines worldwide will find the great lines of thinking and tangible information throughout the book. For professionals in mining and exploration, in inter-governmental and non-governmental organizations (NGOs), the service sector and state administrations, current professional practice is introduced.

sample pages



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INTRODUCTION TO
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(Including Exploration, Mining and Mineral Economics)
Moses A. Olade, PhD., MBA


MINERAL DEPOSITS (3rd Edition)
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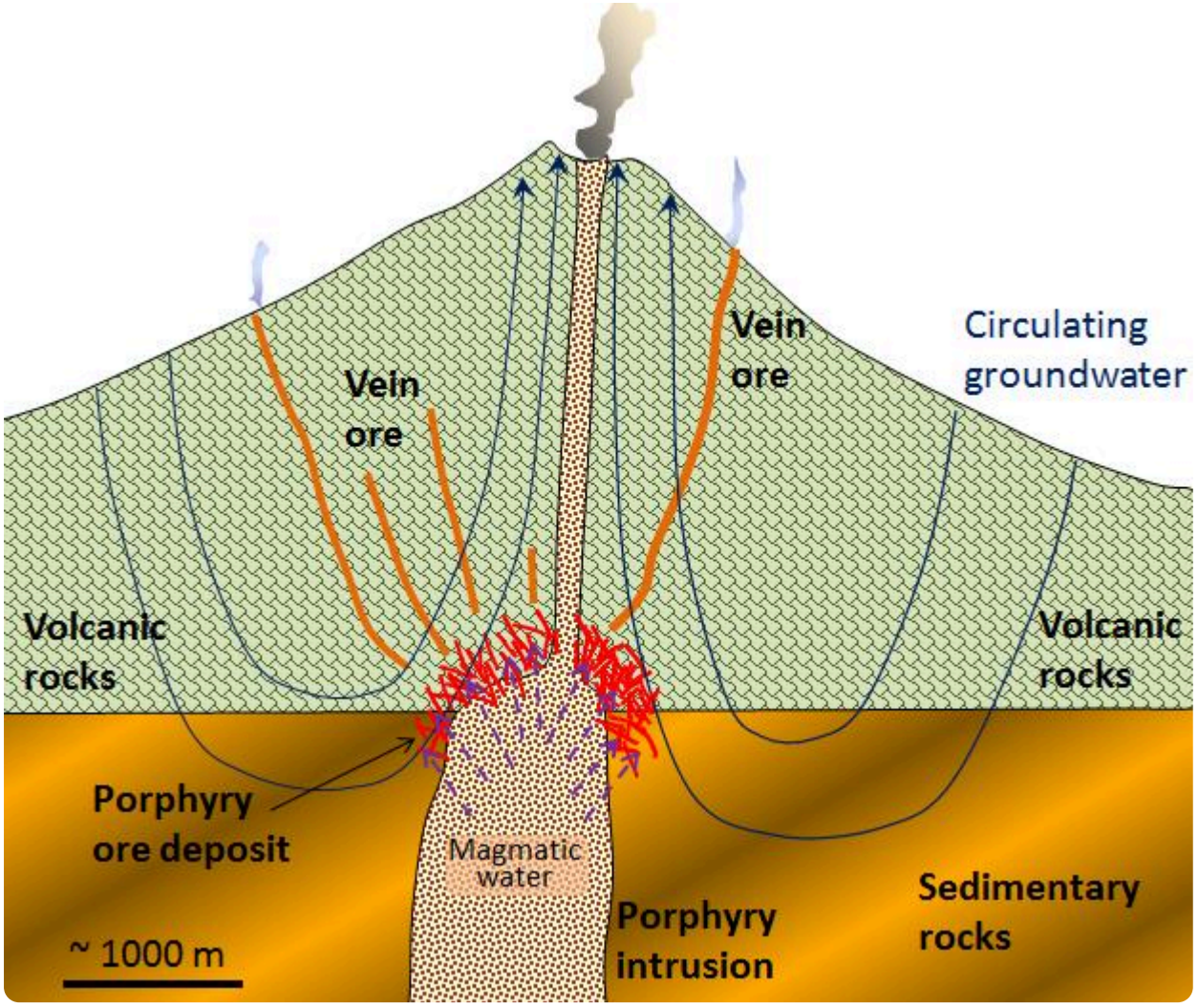
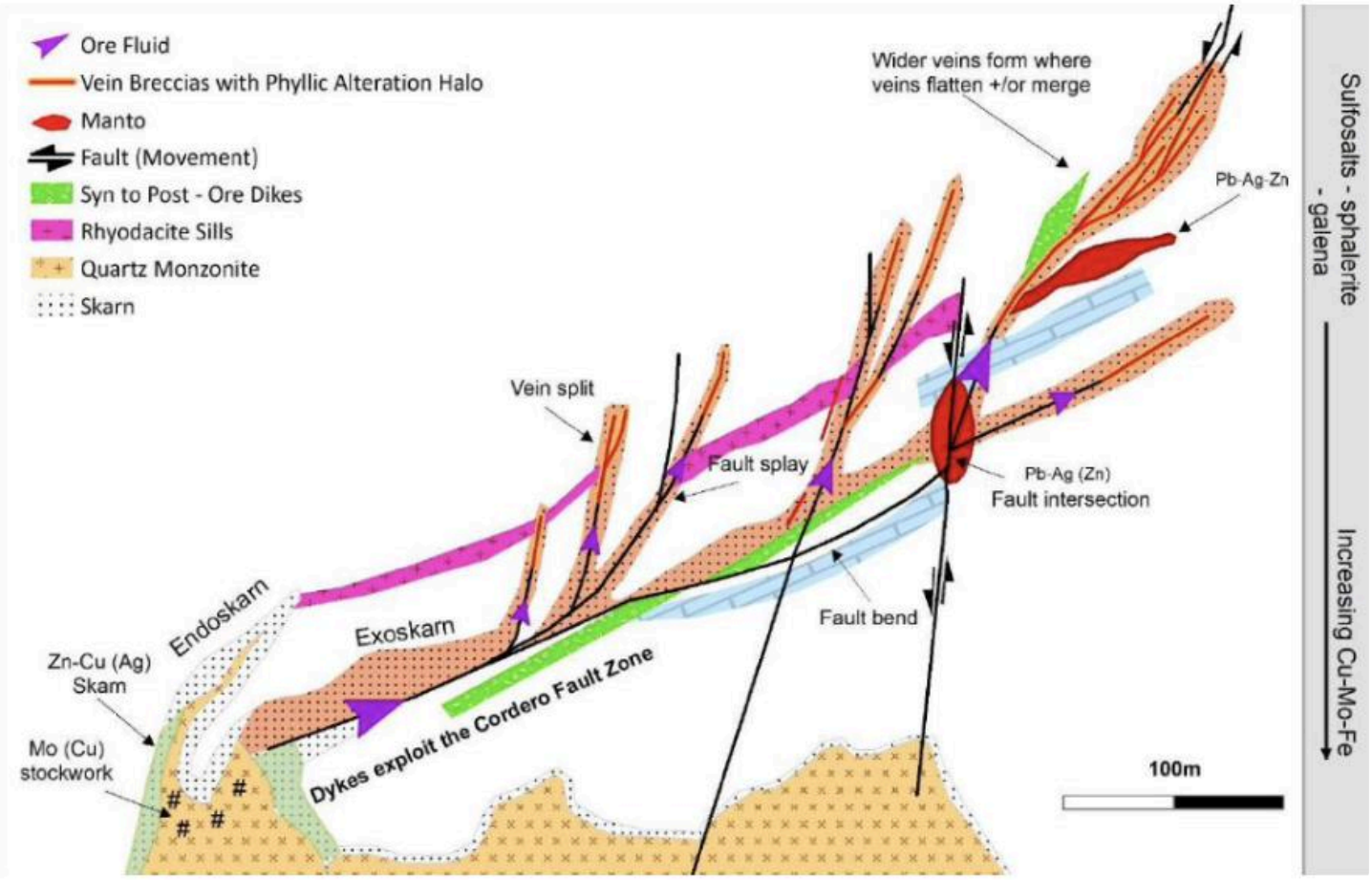
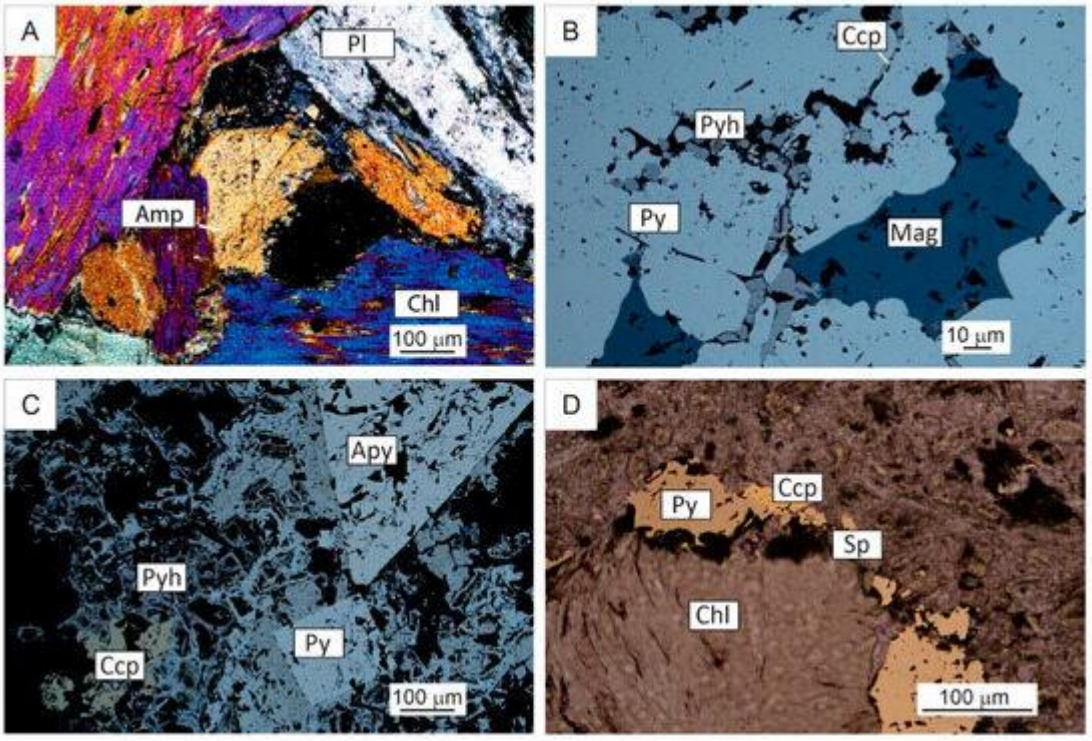

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Figure 3.2_2

Conceptual Mineralisation Model for Cordero





Alternances de couches de chromitite à platinoïdes (niveaux noirs) et d'anorthosite (niveaux blancs), dans le complexe du Bushveld, Afrique du Sud





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